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# Comparison of Hydrus and iStent microinvasive glaucoma surgery implants in combination with phacoemulsification for treatment of open-angle glaucoma: systematic review and network meta-analysis



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|---|---|
| Journal: | *BMJ Open* |
| Manuscript ID | bmjopen-2021-051496 |
| Article Type: | Original research |
| Date Submitted by the Author: | 20-Mar-2021 |
| Complete List of Authors: | Hu, Rongrong; Zhejiang University School of Medicine First Affiliated Hospital, Department of Ophthalmology<br>Guo, Dongyu; Zhejiang University School of Medicine First Affiliated Hospital, Department of Ophthalmology<br>Hong, Nan; Zhejiang University School of Medicine First Affiliated Hospital, Department of Ophthalmology<br>Xuan, Xiuyuan; Zhejiang University School of Medicine First Affiliated Hospital, Department of Ophthalmology<br>Wang, Xiaoyu; Zhejiang University School of Medicine First Affiliated Hospital, Department of Ophthalmology |
| Keywords: | Glaucoma < OPHTHALMOLOGY, Cataract and refractive surgery < OPHTHALMOLOGY, Ophthalmology < SURGERY |
| | |



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# Comparison of Hydrus and iStent microinvasive glaucoma surgery implants in combination with phacoemulsification for treatment of open-angle glaucoma: systematic review and network meta-analysis

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**Word count：** 2908 words

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# ABSTRACT

**Objectives：** To compare the efficacy and safety of two Schlemm's canal-based microinvasive glaucoma surgery (MIGS) devices, the Hydrus Microstent and the iStent Trabecular Bypass combined with phacoemulsification for treatment of open-angle glaucoma.

**Methods:** Literature searches were conducted on PubMed, Web of Science, Cochrane Library and ClinicalTrials.gov to identify relevant randomized controlled trials (RCTs) until Sep 2020. Effects were estimated using the intraocular pressure (IOP) reduction (IOPR), the percentage of IOPR, and the proportion of medication-free patients at follow-up end. Safety was estimated using the proportions of adverse events. The network meta-analysis was conducted within a Bayesian framework using the Markov Chain Monte Carlo method in ADDIS software.

**Results：** Six RCTs comprising 1397 patients were identified. Regarding the absolute value and the percentage of IOPR, the Hydrus and 2-iStent implantation combined with phacoemulsification were significantly more effective than phacoemulsification alone. Rank probability analysis revealed the Hydrus was the best choice to lower IOP. There were no significant differences in the proportion of medication-free patients among groups. The Hydrus and 2-iStent implantation had a higher probability to achieve the medication-free status versus the 1-iStent implantation and phacoemulsification alone. Overall safety profiles were good for each device with the focal peripheral anterior synechiae more frequently reported in Hydrus eyes.




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**Conclusions:** The Hydrus implantation might have a slight advantage over the 1- or 2-iStent implantation in combination with phacoemulsification to treat open-angle glaucoma. Further studies are needed to confirm our findings, including high-quality RCTs to directly compare these MIGS devices.

## Strengths and limitations of this study

- This meta-analysis included the most recent reports.
- We systematically identified and rigorously collected the available evidence to perform a comprehensive comparison of two Schlemm's canal-based microinvasive glaucoma surgery implants.
- The number of included studies is relatively small. Further research is needed to replenish this meta-analysis to offer more convincing conclusion.
- The details of adverse events were not always reported in each study.

**Key words:** open-angle glaucoma, microinvasive glaucoma surgery, trabecular bypass stents, micro-bypass stents, phacoemulsification, intraocular pressure

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## INTRODUCTION

Glaucoma is a leading cause of irreversible vision loss in the world. Although the pathogenesis of glaucoma is not fully clear, the level of intraocular pressure (IOP) has been closely associated with the retinal ganglion cell degeneration in glaucoma.[1] Reduction of IOP is the only proven and generally accepted method to treat glaucoma in slowing disease progression.[1-4] Topical ocular hypotensive medications are usually a first-line choice, however, patient adherence and ocular surface toxicity are major issues with the long-term medical management.[5] When topical medications or other interventions (such as laser) do not achieve adequate IOP reduction, incisional surgery has to be considered. Current mainstream procedures, trabeculectomy and aqueous shunts, while highly effective at IOP reduction, may require quite intense postoperative care for the first 3 months after surgery, and may be associated with several sight-threatening complications.[1,4,6]

In recent years, new devices and procedures have been developed aiming for safer and less invasive IOP reduction compared with conventional glaucoma surgeries.[7] The microinvasive glaucoma surgery (MIGS) devices, placed ab interno via clear corneal incision, shunt aqueous into the Schlemm's canal, the suprachoroidal, or subconjunctival space. These devices can be implanted in combination with cataract surgery to treat mild-to-moderate open-angle glaucoma (OAG).[8] Independent randomized controlled trials (RCTs) have demonstrated that MIGS devices are effective in reducing IOP and medication use with good safety profiles.[9-17]



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In the present network meta-analysis, we aimed to indirectly compare the clinical efficacy and safety of two different Schlemm's canal-based MIGS devices, the Hydrus Microstent (Ivantis, Inc, Irvine, CA) and the iStent Trabecular Bypass device (Glaukos Corporation, San Clemente, CA), combined with phacoemulsification for the treatment of OAG. The Hydrus Microstent is an 8-mm long crescent-shaped open structure, curved to match the shape of Schlemm's canal. The iStent Trabecular Bypass device is a 1.0 mm long L-shaped stent with a snorkel (inlet) on the short side, which resides in the anterior chamber, and an open half-pipe lumen (foot), which resides in Schlemm's canal. The clinical efficacy outcomes include the IOP reduction from baseline to follow-up end, and the proportion of medication-free patients at the end of follow-up. The safety outcomes include the proportion of common adverse events after surgery.

## METHODS

This systematic review was conducted and reported based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) Statement,[18] and the PRISMA network meta-analysis extension statement.[19]

### Patient and public involvement

We used secondary data from peer-reviewed published articles, so there was no direct patient or public involvement in this review.



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**Literature search**

We searched PubMed, Web of Science, Cochrane Library and ClinicalTrials.gov to identify relevant studies in Sep 2020. The following terms were searched in each database: ‘iStent’, ‘trabecular micro-bypass stent’, ‘trabecular bypass’, ‘Hydrus’, ‘Hydrus microstent’, or ‘schlemm canal microstent’ AND ‘cataract’, ‘cataract surgery’, or ‘phacoemulsification’. We did not apply limits for language or date of publication. When titles fit our search terms, abstracts were reviewed to exclude irrelevant studies (e.g., case reports, reviews or experimental studies). We then carefully read all the remaining articles to determine if they contained data that were applicable to the present study. Reference lists of included studies and previous meta-analyses were also manually searched for additional relevant articles.

**Inclusion and exclusion criteria**

We included studies based on the following criteria: (1) The study was a RCT. (2) Hydrus or iStent implantation combined with phacoemulsification was performed. (3) Participants were diagnosed as OAG. (4) Studies had to report baseline and postoperative IOP, and anti-glaucoma medication use. Studies that met any of the following criteria were excluded: (1) review article or case report, (2) duplicate publication, (3) describing a study already included (e.g., subgroup analysis or mid-term report of a large trial, or follow-up report after a trial ends), (4) sufficient information not published (e.g., full text not accessible or full text did not contain raw data).



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**Outcome measures**

For efficacy analysis, the primary outcomes were the IOP reduction (IOPR) and the percentage of IOP reduction (IOPR%). When authors reported the mean and standard deviation (SD) of the IOPR, we used them directly. For studies that only reported absolute values of the IOP at baseline and end-point, the IOPR and the SD of the IOPR ($SD_{IOPR}$) were computed as follows:[20,21]

$$IOPR = IOP_{baseline} - IOP_{end-point},$$

$$SD_{IOPR} = \left(SD_{baseline}^{2} + SD_{end-point}^{2} - SD_{baseline} \times SD_{end-point}\right)^{1/2},$$

then the IOPR% and the SD of the IOPR% ($SD_{IOPR\%}$) were estimated by

$$IOPR\% = \frac{IOPR}{IOP_{baseline}} \times 100\%,$$

$$SD_{IOPR\%} = \frac{SD_{IOPR}}{IOP_{baseline}} \times 100\%.$$

The second outcome measure was the proportion of eyes with complete success, which was defined as targeting end-point IOP (≤21mmHg or study-specific end-points) without anti-glaucoma medication. We assessed safety by considering the proportions of eyes with postoperative adverse events, including device malposition, device obstruction, nonobstructive peripheral anterior synechiae (PAS), hyphema, uveitis/iritis, macular edema, disc hemorrhage, best-corrected visual acuity (BCVA) loss > 2 lines, and elevated IOP ⩾ 10 mmHg over baseline.



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**Risk of bias assessment**

The methodological quality of included studies was evaluated independently by two authors (RH and DG) based on the 6-item modified Jadad scale.[22-24] It evaluates the method of randomization, blinding, withdrawals and drop-outs, inclusion/exclusion criteria, adverse events, and the statistical analysis. According to these criteria, the scale scores range from 0 to 8 points. Studies with a score of 4 to 8 were deemed to be high quality, and studies with a score of 0 to 3 were deemed to be low quality. Any disagreement between raters was resolved through discussion with a third author (XW).

**Data extraction**

The studies' demographic details, participant characteristics, interventions, outcomes and limitations were independently extracted by two authors (RH, DG). If disagreements occurred, XW would check the data again.

**Statistical analyses**

Our analysis classified Hydrus implantation with phacoemulsification, 1-iStent implantation with phacoemulsification, 2-iStent implantation with phacoemulsification, and phacoemulsification alone as separate treatment nodes. The network meta-analysis was performed within a Bayesian framework using the Markov Chain Monte Carlo simulation implemented through the Aggregate Data Drug Information System software (ADDIS, V.1.16.6, Drugis, Groningen, NL).[25] We used a consistency model, which was based on 50,000 iterations for each 4 chains with a burn-in period of the first 20,000 iterations. Convergence was assessed using the Brooks-Gelman-Rubin method.[26] The



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measures of treatment effects were the weighted mean difference (WMD) for continuous outcomes and relative risk (RR) for dichotomous outcomes with 95% credible intervals (95% CrI) based on the Bayesian statistical inference. Consistency analysis could be conducted in the presence of similarity and homogeneity, and on this basis, we could rank the effect of different treatment strategies. The higher ranking means the better the treatment is probably when considering treatment efficacy. It should be noted that even if the difference in the effect size between treatments is small, clinical decisions to choose treatment can still be guided by the probability of treatment rankings.[27,28] Inconsistency analysis was also conducted and similar results were obtained (Supplemental Tables 1-3).

## RESULTS

### Literature search results

The PRISMA flowchart of literature selection in this network meta-analysis is illustrated in Figure.1. In total, 177 articles were retrieved through our initial search. After removal of duplicates and irrelevant studies (e.g., case reports, reviews or experimental studies), 95 articles remained for full-text review. Eighty-two articles were excluded on full-text review as they were not RCTs (43 studies), did not include interventions of interest (11 studies), or did not report outcomes of interest (28 studies). Full-text examinations excluded 7 additional articles as they described a RCT already included [subgroup analysis (3 studies), mid-term report (1 studies), or follow-up report after a trial ends (2 studies)], or did not report sufficient data (1 study). Finally, 6 articles (6 RCTs)[9,11-15] were included in this systematic review and network meta-analysis.



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**Characteristics and outcomes of included studies**

Six studies comprising 1397 eyes with OAG and cataract were included in this meta-analysis. Figure. 2 presents the network of eligible comparisons for the network meta-analysis. All included studies were prospective RCT investigations. The follow-up duration ranged from 12 months to 24 months and the patients' age distributions did not vary significantly among studies. The median sample size was 139 eyes (range 33 - 556). The main characteristics of the included studies are presented in Table 1.

**Methodological quality of included studies**

Figure. 3 presents the results of the risk of bias assessment for each of the included studies. The overall quality scores were good ranging from 5 to 8 (the modified-Jadad score >3). Of the 6 RCTs, appropriate randomization methods were described by 5 studies (83.3%); one study (16.7%) was described as double-blind, 3 studies (50%) were single-blind, and the remaining studies were not blinded (n = 2, 33.3%). Of the included studies, the majority provided an adequate description of withdrawals and dropouts (n = 5, 83.3%), inclusion/exclusion criteria (n = 6, 100%), the method used to assess adverse effects (n = 6, 100%), and the methods of statistical analysis (n = 6, 100%).

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**Table 1.** Study characteristics of included randomized controlled trials

| First author (year) | Location | Design | Interventions (number of eyes) | Age (mean±SD) | Follow-up (month) |
|---|---|---|---|---|---|
| Fea et al. (2010)[9] | Italy | prospective double-masked | 1-iStent (n=12)<br>Control (n=21) | 64.5±3.4<br>64.9±3.1 | 15 |
| Craven et al. (2012)[11] | United States | prospective open-label | 1-iStent (n=98)<br>Control (n=101) | 74±8<br>73±9 | 24 |
| Fernandez-Barrientos et al. (2010)[12] | Spain | prospective open-label | 2-iStent (n=17)<br>Control (n=16) | 75.2±7.2<br>76.7±5.8 | 12 |
| Samuelson et al. (2019)[13] | United States | prospective single-masked | 2-iStent (n=380)<br>Control (n=118) | 69.0±8.2<br>70.1±7.7 | 24 |
| Pfeiffer et al. (2015)[14] | multicenter | prospective single-masked | Hydrus (n=44)<br>Control (n=34) | 72.8±6.6<br>71.5±6.9 | 24 |
| Samuelson et al. (2019)[15] | multicenter | prospective single-masked | Hydrus (n=369)<br>Control (n=187) | 71.1±7.9<br>71.2±7.6 | 24 |

SD, standard deviation; 1-iStent, 1-iStent implantation with phacoemulsification; 2-iStent, 2-iStent implantation with phacoemulsification; Hydrus, Hydrus implantation with phacoemulsification; Control, phacoemulsification alone.



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## Efficacy

Table 2 and 3 present the WMD and 95% CrI of IOPR and IOPR%, respectively for all possible comparisons by the network meta-analysis using the consistency model. In terms of the IOPR, the 2-iStent group (1.88, 95% CrI 0.41 to 3.62 mmHg) and the Hydrus group (2.21, 95% CrI 0.52 to 3.71 mmHg) showed greater IOP reduction than the control group at follow-up end. In terms of the IOPR%, the 1-iStent group (7.65%, 95% CrI 0.66 to 15.45%), the 2-iStent group (7.31%, 95% CrI 1.14 to 14.60%), and the Hydrus group (8.66%, 95% CrI 2.02 to 14.81%) all showed greater IOP reduction than the control group. There was no significant difference in IOPR and IOPR% among the 1-iStent group, the 2-iStent group, and the Hydrus group. Figure. 4-A, 4-B present the rank probability of these groups for the treatment effect in terms of the IOPR and the IOPR%, respectively. Overall, the Hydrus group showed the higher probability to be more effective in lowering IOP than the 1-iStent group and the 2-iStent group.

**Table 2**. Network meta-analysis results in IOP reduction at follow-up end

| Weighted mean difference (95% CrI), mmHg | | | |
|---|---|---|---|
| 1-iStent | | | |
| -0.47 (-2.86, 1.65) | 2-iStent | | |
| -0.81 (-3.02, 1.64) | -0.35 (-2.34, 2.25) | Hydrus | |
| 1.40 (-0.19, 3.06) | 1.88 (0.41, 3.62)* | 2.21 (0.52, 3.71)* | Control |

*Statistical significance.
IOP, intraocular pressure; CrI, credible interval; 1-iStent, 1-iStent implantation with phacoemulsification; 2-iStent, 2-iStent implantation with phacoemulsification; Hydrus, Hydrus implantation with phacoemulsification; Control, phacoemulsification alone.



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**Table 3**. Network meta-analysis results in the percentage of IOP reduction at follow-up end

| Weighted mean difference (95% CrI), % | | | |
|---|---|---|---|
| 1-iStent | | | |
| 0.32 (-10.02, 10.27) | 2-iStent | | |
| -1.08 (-10.31, 9.24) | -1.33 (-9.79, 8.75) | Hydrus | |
| 7.65 (0.66, 15.45)* | 7.31 (1.14, 14.60)* | 8.66 (2.02, 14.81)* | Control |

*Statistical significance.
IOP, intraocular pressure; CrI, credible interval; 1-iStent, 1-iStent implantation with phacoemulsification; 2-iStent, 2-iStent implantation with phacoemulsification; Hydrus, Hydrus implantation with phacoemulsification; Control, phacoemulsification alone.

Table 4 presents the RR and 95% CrI of the proportions of eyes with complete success for all possible comparisons by the network meta-analysis using the consistency model. The 1-iStent group, the 2-iStent group, and the Hydrus group showed a likely positive effect in the proportions of complete success versus the control group, however, the differences were not statistically significant. There was no significant difference in the proportion of complete success among the 1-iStent group, the 2-iStent group, and the Hydrus group. Figure. 4-C presents the rank probability of these groups for treatment effect in terms of the proportion of complete success. The 2-iStent group and the Hydrus group showed the higher probability to achieve the complete success than the 1-iStent group and the control group. The 2-iStent group and the Hydrus group showed the close probability to be the best treatment (0.44 and 0.40, respectively).

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| **Table 4**. Network meta-analysis results in medication free at follow-up end | | | |
|---|---|---|---|
| Relative risk (95% CrI) in proportions of complete success | | | |
| 1-iStent | | | |
| 0.60 (0.01, 10.07) | 2-iStent | | |
| 0.63 (0.04, 16.37) | 1.05 (0.08, 68.13) | Hydrus | |
| 2.65 (0.37, 29.48) | 4.33 (0.79, 145.22) | 4.06 (0.51, 38.62) | Control |

CrI, credible interval; complete success, targeting end-point intraocular pressure (⩽21mmHg or study-specific end-points) without anti-glaucoma medication; 1-iStent, 1-iStent implantation with phacoemulsification; 2-iStent, 2-iStent implantation with phacoemulsification; Hydrus, Hydrus implantation with phacoemulsification; Control, phacoemulsification alone.

**Adverse events**

Main adverse events of included studies are present in Table 5. In summary, device malposition occurred in 3.9% of 1-iStent eyes, 1.4% of 2-iStent eyes, and was not reported in Hydrus eyes. Device obstruction occurred in 3.9% of 1-iStent eyes, 6.0% of 2-iStent eyes, and 3.3% of Hydrus eyes. Focal PAS, which was considered as nonobstruction, occurred in 1.7% of 2-iStent eyes, 15.3% of Hydrus eyes, and was not reported in 1-iStent eyes.

There were few sight-threatening complications in included studies. At follow-up end, 2.5% of 2-iStent eyes and 1.2% of Hydrus eyes had lost >2 lines compared with preoperative BCVA, however, the proportions were numerically less in treatment eyes than control eyes (3.7% and 1.7%, respectively in 2-iStent and Hydrus studies). Macular edema occurred in 0.8% of 1-iStent eyes and 2.4% of Hydrus eyes, which were also less in treatment eyes than control eyes (1.4% and 2.5%, respectively in 1-iStent and Hydrus studies). Hyphema, disc hemorrhage, and IOP elevation ⩾10 mmHg over baseline occurred at a rate of 2% or less in all treatment eyes of included studies. A higher



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proportion of postoperative uveitis/iritis was observed in 2-iStent eyes and Hydrus eyes versus control eyes (5.5% vs. 3.7% in 2-iStent studies; 5.0% vs. 2.9% in Hydrus studies), while a lower proportion of uveitis/iritis was reported in 1-iStent eyes versus control eyes (0.8% vs. 4.3%).

**Table 5**. Main adverse events after Hydrus and iStent implantation combined with phacoemulsification

| Adverse events | 1-iStent | 1-iStent Control | 2-iStent | 2-iStent Control | Hydrus | Hydrus Control |
|---|---|---|---|---|---|---|
| Number of eyes | 128 | 138 | 403 | 135 | 417 | 236 |
| Device malposition | 3.9% | NA | 1.4% | NA | NR | NA |
| Device obstruction | 3.9% | NA | 6.0% | NA | 3.3% | NA |
| Focal PAS, nonobstructive | NR | NR | 1.7% | NR | 15.3% | 2.1% |
| Hyphema | NR | NR | 2.0% | NR | 0.4% | 0.4% |
| Uveitis/iritis | 0.8% | 4.3% | 5.5% | 3.7% | 5.0% | 2.9% |
| Macular edema | 0.8% | 1.4% | NR | NR | 2.4% | 2.5% |
| Disc hemorrhage | 0.8% | 2.2% | NR | NR | 0.2% | NR |
| BCVA loss > 2 lines | NR | NR | 2.5% | 3.7% | 1.2% | 1.7% |
| Elevated IOP ≥ 10 mmHg over baseline | NR | NR | 2.0% | 0.7% | 0.9% | 3.0% |

NA, not applicable; NR, not reported; PAS, peripheral anterior synechiae; BCVA, best-corrected visual acuity; IOP, intraocular pressure; 1-iStent, 1-iStent implantation with phacoemulsification; 2-iStent, 2-iStent implantation with phacoemulsification; Hydrus, Hydrus implantation with phacoemulsification.



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## DISCUSSION

To our knowledge, this is the first study to indirectly compare the efficacy and safety of two commercially available Schlemm's canal-based MIGS devices for use in glaucoma management with concurrent cataract surgery. Based on the network meta-analysis, the Hydrus and 2-iStent implantation combined with phacoemulsification achieved statistically greater reductions in IOP than phacoemulsification alone, in terms of either the absolute IOP reduction values or the percentage. Although no significant difference was found among the three groups, the Hydrus device may provide better IOP reduction effect than the 1- and 2-iStent device based on the rank probability of the existing data. In terms of the proportion of complete success, no statistically significant difference was found among the 1-iStent group, the 2-iStent group, the Hydrus group, and the phacoemulsification alone group, while the rank probability analysis shows the Hydrus and the 2-iStent implantation may achieve the complete success at a higher probability versus the 1-iStent implantation and phacoemulsification alone.

Previous laboratory studies compared the Hydrus and the iStent devices using human cadaveric anterior segment perfusion models.[29,30] Hays et al showed the Hydrus increased the aqueous outflow facility by a greater percentage (73% vs. 34%) and at a wider range of perfusion pressures (20 to 50 mm Hg vs. 40 mm Hg) than the 2-iStent.[29] Similar results were shown in a recent randomized paired comparison study by Toris et al, in which Hydrus eyes had higher increase in the outflow facility than 2-iStent eyes (75% vs. 39%).[30] These in-vitro laboratory findings support our meta-analysis results that the Hydrus device may lower the IOP better than the 2-iStent device. Furthermore, the



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negative correlation between the Schlemm's canal area and the IOP has been observed in vivo in OAG eyes, and normal eyes have larger Schlemm's canal area than OAG eyes.[31] Trabecular meshwork stretch is also one of mechanisms of the pilocarpine treatment effect.[32] It is therefore likely that the trabecular meshwork stretch and Schlemm's canal dilation by the 8-mm scaffold segment of the Hydrus help to provide additional benefits to lower the outflow resistance to collector channels, and accordingly achieve greater reduction in IOP versus the iStent.[29,30]

Our results are also in line with the conclusion of a recent prospective, multicenter RCT that compared the Hydrus and the 2-iStent for standalone treatment.[33] One hundred fifty-two eyes from 152 OAG patients were randomized to either the Hydrus implantation or the 2-iStent implantation and followed for 12 months. Both groups had significantly reduced medication use at 12 months compared with the baseline, and the reduction in the Hydrus group was significantly greater by 0.6 medication than the 2-iStent group. More Hydrus patients were medication free at the follow-up end (46.6% vs. 24.0%). While it is difficult to directly compare the present meta-analysis results of combined surgery with the standalone treatment results, both conclusions suggest the Hydrus might be more effective to treat OAG versus the 2-iStent.

The overall safety profiles of the treatment groups were favorable and few serious ocular adverse events were reported in included RCTs. Device obstruction, which may be caused by pigment, iris tissue, or hyphema, occurred in either device (3.9% of 1-iStent eyes, 6.0% of 2-iStent eyes, and 3.3% of Hydrus eyes). Device malposition was observed



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in 3.9% of 1-iStent eyes and 1.4% of 2-iStent eyes, while not reported in the Hydrus eyes. A likely explanation is the Hydrus was designed with a longer segment residing within the lumen of Schlemm's canal versus the iStent, which allows a better positioning stability. PAS was more frequently observed in the Hydrus eyes versus the 1-iStent and 2-iStent eyes (15.3%, 0%, and 1.7%, respectively). Our result is consistent with the standalone Hydrus and 2-iStent comparison study, which indicates that the PAS formation was more common in the Hydrus group.[33] It is noteworthy that subgroup analyses of included RCTs did not show the significant difference in the IOP reduction between eyes with PAS and without PAS.[14,15] Therefore, most of the PAS might not be obstructive in eyes treated with Hydrus implantation with phacoemulsification. Nevertheless, caution should be taken for the PAS as it may eventually result in the device obstruction in the long term.

Although the results of this meta-analysis may be important for treatment considerations, there are several imitations that should be mentioned. First, we included prospective high-quality RCTs, however, the number of studies is relatively small. Further research is still needed to replenish this meta-analysis to offer more convincing conclusion. Second, the sample size of included studies varied ranging from 33 to 556 eyes, which may potentially bias the results. Third, the details of adverse events were not always reported in each study, and owing to the limited data, we did not perform statistical analysis to compare the difference in treatment safety among groups.



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## CONCLUSION

Based on the existing data at present, our network meta-analysis results indicate that the Hydrus device might be more effective in combination with phacoemulsification to treat OAG versus the 1- or 2-iStent device, and both devices have overall good safety profiles. Even with the limitations, we believe that the results of the present meta-analysis are credible and clinically useful for treatment considerations in glaucoma management, particularly for patients with mild-to-moderate glaucoma and cataract. Further prospective, large-scale RCTs that directly compare different MIGS devices combined with cataract surgery are needed to confirm our results.

**Acknowledgements** The authors thank the researchers whose studies were included in this network meta-analysis and provided useful data to us.

**Contributors** All authors made a substantial contribution to this work. RH: study conception, study design, data collection, data analysis, manuscript writing and final manuscript approval; DG: study design, data collection and critical manuscript revision; NH: statistical analysis; XX: statistical analysis; XW: study conception, study design, critical manuscript revision and final manuscript approval.

**Funding** This study was supported by the National Natural Science Foundation of China Grant 81900854, the Zhejiang Provincial Natural Science Foundation of China Grant LQ17H120003, and the Health and Family Planning Commission of Zhejiang Province Grant 2016KYA089.



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**Competing interests** None declared.

**Patient consent** Not required.

**Provenance and peer review** Not commissioned; externally peer reviewed.

**Data sharing statement** Extra data are available from the first author on request.

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## Figure legends

**Figure. 1.** Flow chart of study selection process. RCT, randomized controlled trial.

**Figure. 2.** Network graph of all treatment comparisons of included studies. Each node represents a treatment strategy. Lines represent direct comparisons within the randomized controlled trials.

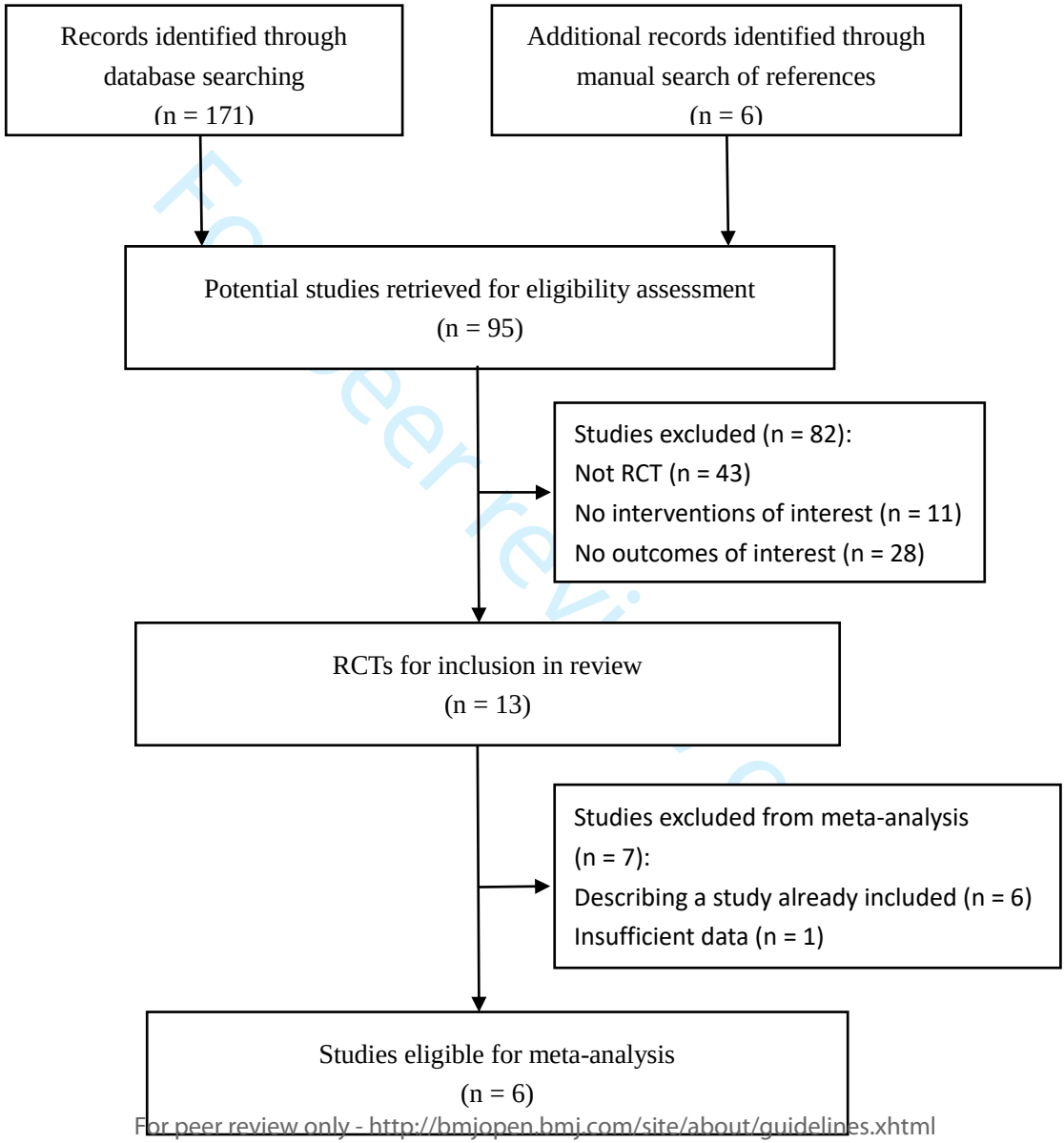
**Figure. 3.** Modified-Jadad scores for risk of bias assessment. NR, not reported; NA, not applicable.

**Figure. 4.** Rank probability of treatment. The rank probability was shown according to absolute value of intraocular pressure (IOP) reduction (A), the percentage of IOP reduction (B), and the proportion of complete success (C). Rank 1 is best and Rank N is worst.

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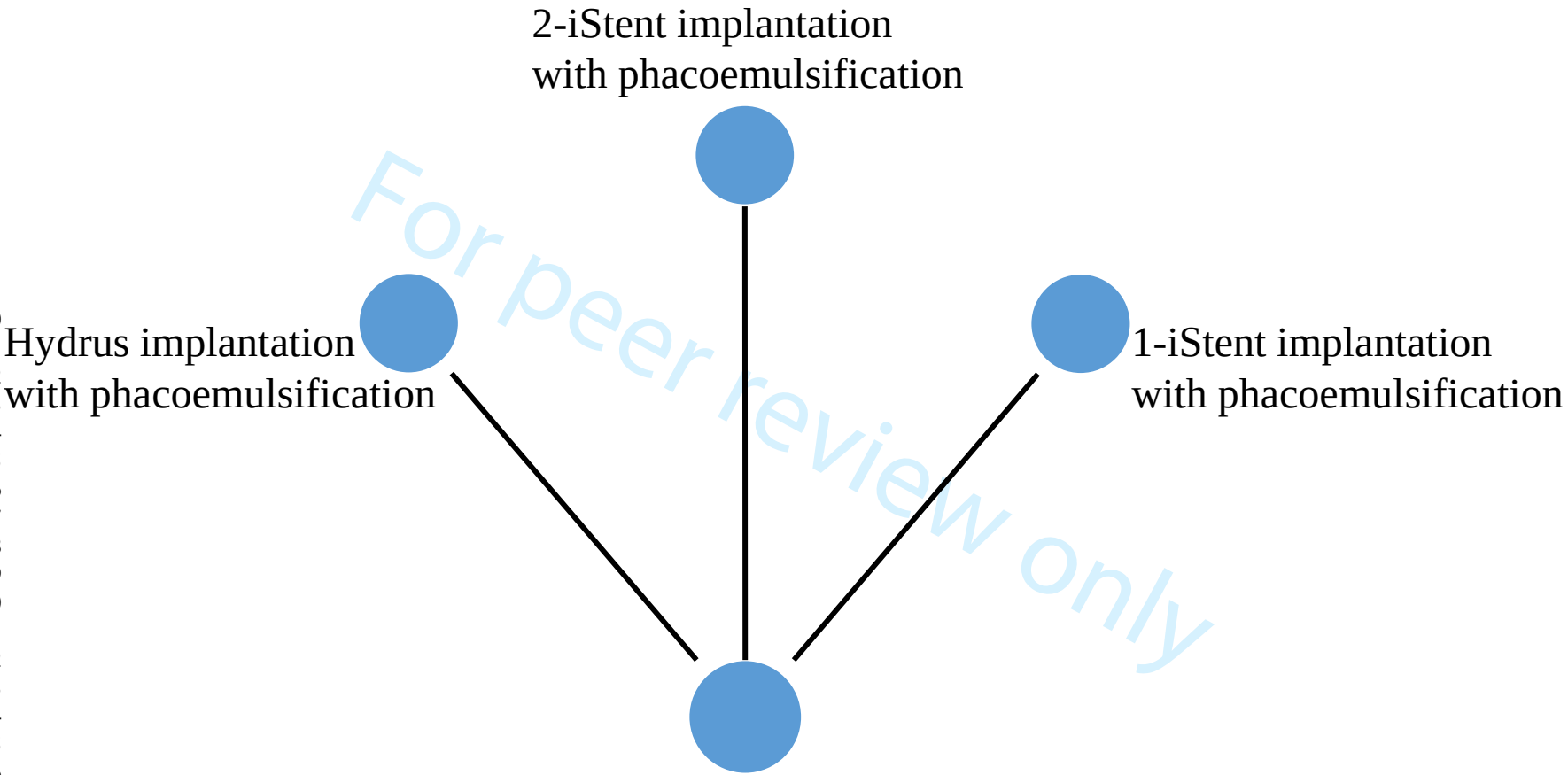


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Fig. 1



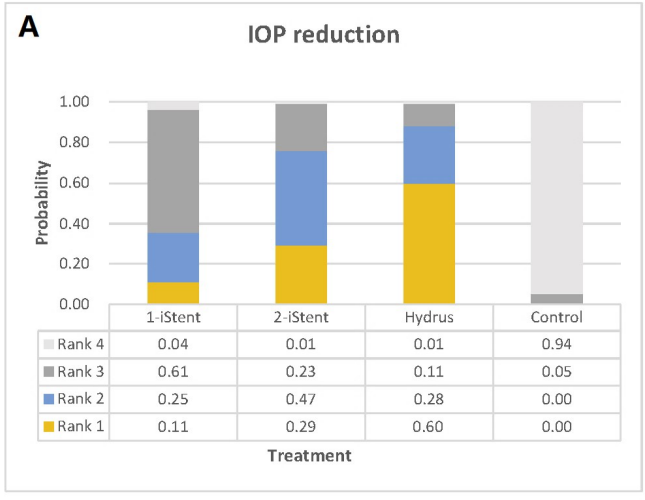

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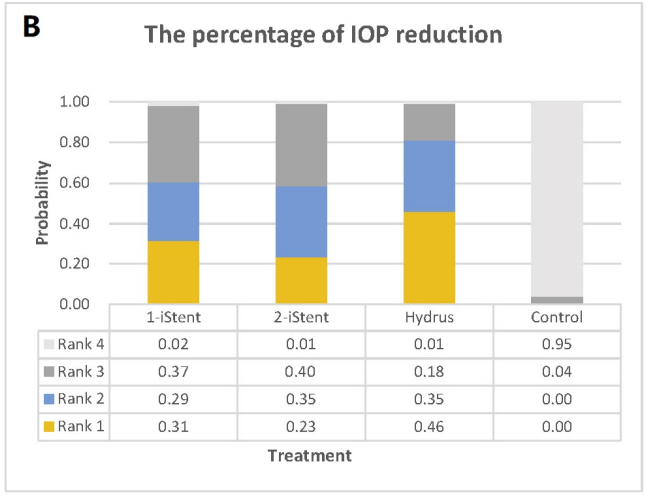
Fig. 2

| | Randomized study | Randomization method appropriate | Double blind study | Blinding method appropriate | Description of withdrawals and dropouts | Description of inclusion/exclusion criteria | Method to assess adverse effects described | Methods of statistical analysis described | TOTAL |
|---|---|---|---|---|---|---|---|---|---|
| Fea et al. (2010)[9] | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 8 |
| Craven et al. (2012)[11] | 1 | NR | 0 | NA | 1 | 1 | 1 | 1 | 5 |
| Fernandez-Barrientos et al. (2010) [12] | 1 | 1 | 0 | NA | NR | 1 | 1 | 1 | 5 |
| Samuelson et al. (2019)[13] | 1 | 1 | 0 | NA | 1 | 1 | 1 | 1 | 6 |
| Pfeiffer et al. (2015)[14] | 1 | 1 | 0 | NA | 1 | 1 | 1 | 1 | 6 |
| Samuelson et al. (2019)[15] | 1 | 1 | 0 | NA | 1 | 1 | 1 | 1 | 6 |

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Fig. 3


A
IOP reduction
1.00
0.80
0.60
0.40
0.20
0.00
Probability
1-iStent 2-iStent Hydrus Control
Rank 4 0.04 0.01 0.01 0.94
Rank 3 0.61 0.23 0.11 0.05
Rank 2 0.25 0.47 0.28 0.00
Rank 1 0.11 0.29 0.60 0.00
Treatment


B
The percentage of IOP reduction
1.00
0.80
0.60
0.40
0.20
0.00
Probability
1-iStent 2-iStent Hydrus Control
Rank 4 0.02 0.01 0.01 0.95
Rank 3 0.37 0.40 0.18 0.04
Rank 2 0.29 0.35 0.35 0.00
Rank 1 0.31 0.23 0.46 0.00
Treatment

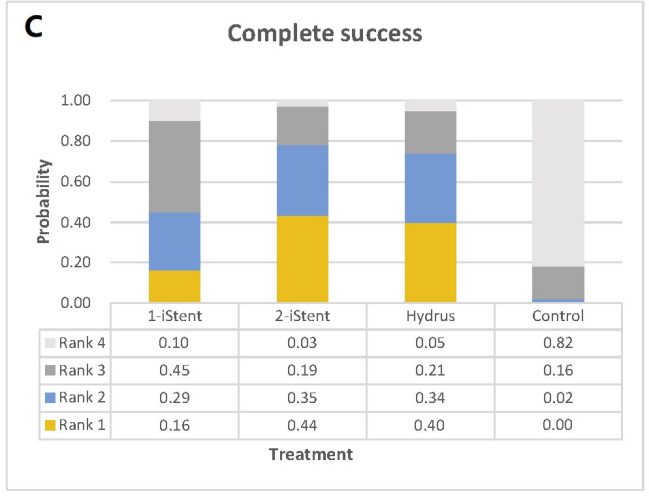

C
Complete success
1.00
0.80
0.60
0.40
0.20
0.00
Probability
1-iStent 2-iStent Hydrus Control
Rank 4 0.10 0.03 0.05 0.82
Rank 3 0.45 0.19 0.21 0.16
Rank 2 0.29 0.35 0.34 0.02
Rank 1 0.16 0.44 0.40 0.00
Treatment

Fig. 4

**Supplemental Table 1**. Network meta-analysis results in IOP reduction at follow-up end (Inconsistency Model).

| Weighted mean difference (95% CrI), mmHg | | | |
|---|---|---|---|
| 1-iStent | | | |
| -0.46 (-2.81, 1.75) | 2-iStent | | |
| -0.81 (-2.98, 1.61) | -0.33 (-2.47, 2.13) | Hydrus | |
| 1.44 (-0.13, 3.19) | 1.90 (0.43, 3.67)* | 2.23 (0.62, 3.77)* | Control |

*Statistical significance.

IOP, intraocular pressure; CrI, credible interval; 1-iStent, 1-iStent implantation with phacoemulsification; 2-iStent, 2-iStent implantation with phacoemulsification; Hydrus, Hydrus implantation with phacoemulsification; Control, phacoemulsification alone.

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**Supplemental Table 2**. Network meta-analysis results in the percentage of IOP reduction at follow-up end (Inconsistency Model).

| Weighted mean difference (95% CrI), % | | | |
|---|---|---|---|
| 1-iStent | | | |
| 0.62 (-9.89, 9.96) | 2-iStent | | |
| -1.03 (-10.66, 9.24) | -1.62 (-9.84, 8.69) | Hydrus | |
| 7.82 (0.21, 15.43)* | 7.19 (1.27, 14.75)* | 8.80 (2.10, 14.92)* | Control |

*Statistical significance.

IOP, intraocular pressure; CrI, credible interval; 1-iStent, 1-iStent implantation with phacoemulsification; 2-iStent, 2-iStent implantation with phacoemulsification; Hydrus, Hydrus implantation with phacoemulsification; Control, phacoemulsification alone.

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**Supplemental Table 3**. Network meta-analysis results in medication free at follow-up end (Inconsistency Model).

| Relative risk (95% CrI) in proportions of complete success | | | |
|---|---|---|---|
| 1-iStent | | | |
| 0.59 (0.01, 10.61) | 2-iStent | | |
| 0.66 (0.04, 18.08) | 1.15 (0.08, 84.20) | Hydrus | |
| 2.63 (0.36, 31.82) | 4.44 (0.76, 139.89) | 3.98 (0.45, 36.31) | Control |

CrI, credible interval; complete success, targeting end-point intraocular pressure (⩽21mmHg or study-specific end-points) without anti-glaucoma medication; 1-iStent, 1-iStent implantation with phacoemulsification; 2-iStent, 2-iStent implantation with phacoemulsification; Hydrus, Hydrus implantation with phacoemulsification; Control, phacoemulsification alone.

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# Appendix 1. PRISMA NMA Checklist of items to include when reporting a systematic review involving a network meta-analysis

| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| **TITLE** | | | |
| Title | 1 | Identify the report as a systematic review incorporating a network meta-analysis (or related form of meta-analysis). | P1 |
| **ABSTRACT** | | | |
| Structured summary | 2 | Provide a structured summary including, as applicable:<br>Background: main objectives<br>Methods: data sources; study eligibility criteria, participants, and interventions; study appraisal; and synthesis methods, such as network meta-analysis.<br>Results: number of studies and participants identified; summary estimates with corresponding confidence/credible intervals; treatment rankings may also be discussed. Authors may choose to summarize pairwise comparisons against a chosen treatment included in their analyses for brevity.<br>Discussion/Conclusions: limitations; conclusions and implications of findings.<br>Other: primary source of funding; systematic review registration number with registry name. | P2-3 |
| **INTRODUCTION** | | | |
| Rationale | 3 | Describe the rationale for the review in the context of what is already known, including mention of why a network meta-analysis has been conducted. | P4-5 |
| Objectives | 4 | Provide an explicit statement of questions being addressed, with reference to participants, interventions, comparisons, outcomes, and study design (PICOS). | P5 |
| **METHODS** | | | |
| Protocol and registration | 5 | Indicate whether a review protocol exists and if and where it can be accessed (e.g., Web address); and, if available, provide registration information, including registration number. | P5 |
| Eligibility criteria | 6 | Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale. Clearly describe eligible treatments included in the treatment network, and note whether any have been clustered or merged into the same node (with justification). | P6 |
| Information sources | 7 | Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched. | P6 |
| Search | 8 | Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated. | P6 |
| Study selection | 9 | State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis). | Figure 1 |
| Data collection process | 10 | Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators. | P8 |

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| | | | |
|---|---|---|---|
| Data items | 11 | List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made. | P7 |
| **Geometry of the network** | S1 | Describe methods used to explore the geometry of the treatment network under study and potential biases related to it. This should include how the evidence base has been graphically summarized for presentation, and what characteristics were compiled and used to describe the evidence base to readers. | P8-9 |
| Risk of bias in individual studies | 12 | Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis. | P8 |
| Summary measures | 13 | State the principal summary measures (e.g., risk ratio, difference in means). Also describe the use of additional summary measures assessed, such as treatment rankings and surface under the cumulative ranking curve (SUCRA) values, as well as modified approaches used to present summary findings from meta-analyses. | P9 |
| Planned methods of analysis | 14 | Describe the methods of handling data and combining results of studies for each network meta-analysis. This should include, but not be limited to:<br>• Handling of multi-arm trials;<br>• Selection of variance structure;<br>• Selection of prior distributions in Bayesian analyses; and<br>• Assessment of model fit. | P8-P9 |
| **Assessment of Inconsistency** | S2 | Describe the statistical methods used to evaluate the agreement of direct and indirect evidence in the treatment network(s) studied. Describe efforts taken to address its presence when found. | P9 |
| Risk of bias across studies | 15 | Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies). | P8 |
| Additional analyses | 16 | Describe methods of additional analyses if done, indicating which were pre-specified. This may include, but not be limited to, the following:<br>• Sensitivity or subgroup analyses;<br>• Meta-regression analyses;<br>• Alternative formulations of the treatment network; and<br>• Use of alternative prior distributions for Bayesian analyses (if applicable). | P8-9 |
| **RESULTS** | | | |
| Study selection | 17 | Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram. | P9, Figure 1 |
| **Presentation of network structure** | S3 | Provide a network graph of the included studies to enable visualization of the geometry of the treatment network. | Figure 2 |
| **Summary of network geometry** | S4 | Provide a brief overview of characteristics of the treatment network. This may include commentary on the abundance of trials and randomized patients for the different interventions and pairwise comparisons in the network, gaps of evidence in the treatment network, and potential biases reflected by the network structure. | P10, Table1 |
| Study characteristics | 18 | For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations. | P10, Table1 |
| Risk of bias within studies | 19 | Present data on risk of bias of each study and, if available, any outcome level assessment. | Figure 3 |
| Results of individual studies | 20 | For all outcomes considered (benefits or harms), present, for each study: 1) simple summary data for each intervention group, and 2) effect estimates and confidence intervals. Modified approaches may be needed to deal with information from larger networks. | P12-15 |
| Synthesis of results | 21 | Present results of each meta-analysis done, including confidence/credible intervals. In larger networks, authors may focus on comparisons versus a particular comparator (e.g. placebo or standard care), with full findings presented in an appendix. League tables and forest plots may be considered to summarize pairwise | P12-15<br>Figure 4 |

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| | | | |
|---|---|---|---|
| | | comparisons. If additional summary measures were explored (such as treatment rankings), these should also be presented. | |
| **Exploration for inconsistency** | S5 | Describe results from investigations of inconsistency. This may include such information as measures of model fit to compare consistency and inconsistency models, P values from statistical tests, or summary of inconsistency estimates from different parts of the treatment network. | Supplemental tables 1-3 |
| Risk of bias across studies | 22 | Present results of any assessment of risk of bias across studies for the evidence base being studied. | Figure 3 |
| Additional analysis | 23 | Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression analyses, alternative network geometries studied, alternative choice of prior distributions for Bayesian analyses, and so forth). | NA |
| **DISCUSSION** | | | |
| Summary of evidence | 24 | Summarize the main findings, including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy-makers). | P16 |
| Limitations | 25 | Discuss limitations at study and outcome level (e.g., risk of bias), and at review level (e.g., incomplete retrieval of identified research, reporting bias). Comment on the validity of the assumptions, such as transitivity and consistency. Comment on any concerns regarding network geometry (e.g., avoidance of certain comparisons). | P18 |
| Conclusions | 26 | Provide a general interpretation of the results in the context of other evidence, and implications for future research. | P19 |
| **FUNDING** | | | |
| Funding | 27 | Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review. This should also include information regarding whether funding has been received from manufacturers of treatments in the network and/or whether some of the authors are content experts with professional conflicts of interest that could affect use of treatments in the network. | P19 |

PICOS = population, intervention, comparators, outcomes, study design.
† Authors may wish to plan for use of appendices to present all relevant information in full detail for items in this section.

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# Comparison of Hydrus and iStent microinvasive glaucoma surgery implants in combination with phacoemulsification for treatment of open-angle glaucoma: systematic review and network meta-analysis

| | |
|---|---|
| Journal: | *BMJ Open* |
| Manuscript ID | bmjopen-2021-051496.R1 |
| Article Type: | Original research |
| Date Submitted by the Author: | 15-Mar-2022 |
| Complete List of Authors: | Hu, Rongrong; Zhejiang University School of Medicine First Affiliated Hospital, Department of Ophthalmology<br>Guo, Dongyu; Zhejiang University School of Medicine First Affiliated Hospital, Department of Ophthalmology<br>Hong, Nan; Zhejiang University School of Medicine First Affiliated Hospital, Department of Ophthalmology<br>Xuan, Xiuyuan; Zhejiang University School of Medicine First Affiliated Hospital, Department of Ophthalmology<br>Wang, Xiaoyu; Zhejiang University School of Medicine First Affiliated Hospital, Department of Ophthalmology |
| Primary Subject Heading: | Ophthalmology |
| Secondary Subject Heading: | Surgery |
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# Comparison of Hydrus and iStent microinvasive glaucoma surgery implants in combination with phacoemulsification for treatment of open-angle glaucoma: systematic review and network meta-analysis

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**Word count：** 4994 words

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# ABSTRACT

**Objectives：** To compare the efficacy and safety of two Schlemm's canal-based microinvasive glaucoma surgery (MIGS) devices, the Hydrus Microstent and the iStent Trabecular Bypass combined with phacoemulsification for treatment of open-angle glaucoma.

**Design:** Systematic review and network meta-analysis.

**Methods:** Literature searches were conducted on PubMed, Web of Science, Cochrane Library and ClinicalTrials.gov to identify randomized controlled trials (RCTs) assessing the Hydrus or the iStent implantation combined with phacoemulsification for treatment of open-angle glaucoma until Sep 2020. Risk of bias was assessed using a 6-item modified Jadad scale. Effects were estimated using the intraocular pressure (IOP) reduction (IOPR), the percentage of IOPR, and the proportion of medication-free patients at follow-up end. Safety was estimated using the proportions of adverse events. The network meta-analysis was conducted within a Bayesian framework using the Markov Chain Monte Carlo method in ADDIS software.

**Results：** Six prospective RCTs comprising 1397 patients were identified. Regarding the absolute value and the percentage of IOPR, the Hydrus and 2-iStent implantation combined with phacoemulsification were significantly more effective than phacoemulsification alone. Rank probability analysis revealed the Hydrus might be the best choice to lower IOP. There was no significant difference in the proportion of medication-free patients among groups. The Hydrus and 2-iStent implantation had a higher probability to achieve the medication-free status versus the 1-iStent implantation




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and phacoemulsification alone. Overall safety profiles were good for each device with the focal peripheral anterior synechiae more frequently reported in Hydrus eyes.

**Conclusions:** The Hydrus implantation may have a slight advantage over the 1- or 2-iStent implantation in combination with phacoemulsification to treat open-angle glaucoma. Our findings might be of some uncertainty due to the limited included data. Further studies are needed to investigate whether our findings are robust, including high-quality RCTs to directly compare these MIGS devices.

## Strengths and limitations of this study

- This meta-analysis included the most recent reports.
- We systematically identified and rigorously collected the available evidence to perform a comprehensive comparison of two Schlemm's canal-based microinvasive glaucoma surgery implants.
- The number of included studies is relatively small. Further research is needed to replenish this meta-analysis to offer more convincing conclusion.
- The details of adverse events were not always reported in each study.

**Key words:** open-angle glaucoma, microinvasive glaucoma surgery, trabecular bypass stents, micro-bypass stents, phacoemulsification, intraocular pressure



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## INTRODUCTION

Glaucoma is a leading cause of irreversible vision loss in the world. Although the pathogenesis of glaucoma is not fully clear, the level of intraocular pressure (IOP) has been closely associated with the retinal ganglion cell degeneration in glaucoma.[1] Reduction of IOP is the only proven and generally accepted method to treat glaucoma in slowing disease progression.[1-4] Topical ocular hypotensive medications are usually a first-line choice, however, patient adherence and ocular surface toxicity are major issues with the long-term medical management.[5] When topical medications or other interventions (such as laser) do not achieve adequate IOP reduction, incisional surgery has to be considered. Current mainstream procedures, trabeculectomy and aqueous shunts, while highly effective at IOP reduction, may require quite intense postoperative care for the first 3 months after surgery, and may be associated with several sight-threatening complications.[1,4,6]

In recent years, new devices and procedures have been developed aiming for safer and less invasive IOP reduction compared with conventional glaucoma surgeries.[7] The microinvasive glaucoma surgery (MIGS) devices, placed ab interno via clear corneal incision, shunt aqueous humor into the Schlemm's canal, the suprachoroidal, or subconjunctival space. These devices can be implanted in combination with cataract surgery to treat mild-to-moderate open-angle glaucoma (OAG).[8] Independent randomized controlled trials (RCTs) have demonstrated that MIGS devices are effective in reducing IOP and medication use with good safety profiles.[9-17]



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In the present network meta-analysis, we aimed to indirectly compare the clinical efficacy and safety of two different Schlemm's canal-based MIGS devices, the Hydrus Microstent (Ivantis, Inc, Irvine, CA) and the iStent Trabecular Bypass device (Glaukos Corporation, San Clemente, CA), combined with phacoemulsification for the treatment of OAG. The Hydrus Microstent is an 8-mm long crescent-shaped open structure, curved to match the shape of Schlemm's canal. The first-generation iStent Trabecular Bypass device is a 1.0 mm long L-shaped stent with a snorkel (inlet) on the short side, which resides in the anterior chamber, and an open half-pipe lumen (foot), which resides in Schlemm's canal. The iStent inject, which is the second generation, preloads two 0.4-mm long, 0.3-mm wide plug-shaped implants with a central opening in its injector for easier insertion. The two-implant design of iStent inject may also reduce the impact of single stent blockage.[18] The clinical efficacy outcomes include the IOP reduction from baseline to follow-up end, and the proportion of medication-free patients at the end of follow-up. The safety outcomes include the proportion of common adverse events after surgery.

## METHODS

This systematic review was conducted and reported based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) Statement,[19] and the PRISMA network meta-analysis extension statement.[20]

### Patient and public involvement

This systematic review was conceptualized and developed due to the unmet need of mild-to-moderate OAG patients to select an effective and safe treatment for IOP reduction. We



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used secondary data from peer-reviewed published articles. Even though there was no direct patient or public involvement in this review, the study compared two Schlemm's canal-based MIGS devices comprehensively based on the existing patient data and aim to aid clinicians to properly consult and treat future patients.

**Literature search**

We searched PubMed, Web of Science, Cochrane Library and ClinicalTrials.gov to identify relevant studies in Sep 2020. The following terms were searched in each database: 'iStent', 'trabecular micro-bypass stent', 'trabecular bypass', 'Hydrus', 'Hydrus microstent', or 'schlemm canal microstent' AND 'cataract', 'cataract surgery', or 'phacoemulsification' (see online supplementary appendix 1 for further details on search strategy). We did not apply limits for language or date of publication. When titles fit our search terms, abstracts were reviewed to exclude irrelevant studies (e.g., case reports, reviews or experimental studies). We then carefully read all the remaining articles to determine if they contained data that were applicable to the present study. Reference lists of included studies and previous meta-analyses were also manually searched for additional relevant articles.

**Inclusion and exclusion criteria**

We included studies based on the following criteria: (1) The study was a RCT. (2) Hydrus or iStent implantation combined with phacoemulsification was performed. (3) Participants were diagnosed as OAG. (4) Studies had to report baseline and postoperative IOP, and anti-glaucoma medication use. Studies that met any of the following criteria



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were excluded: (1) review article or case report, (2) duplicate publication, (3) describing a study already included (e.g., subgroup analysis or mid-term report of a large trial, or follow-up report after a trial ends), (4) sufficient information not published (e.g., full text not accessible or full text did not contain raw data).

**Outcome measures**

For efficacy analysis, the primary outcomes were the IOP reduction (IOPR) and the percentage of IOP reduction (IOPR%). When authors reported the mean and standard deviation (SD) of the IOPR, we used them directly. For studies that only reported absolute values of the IOP at baseline and end-point, the IOPR and the SD of the IOPR ($SD_{IOPR}$) were computed as follows:[21,22]

$$IOPR = IOP_{baseline} - IOP_{end-point},$$

$$SD_{IOPR} = \left(SD_{baseline}^{2} + SD_{end-point}^{2} - SD_{baseline} \times SD_{end-point}\right)^{1/2},$$

then the IOPR% and the SD of the IOPR% ($SD_{IOPR\%}$) were estimated by

$$IOPR\% = \frac{IOPR}{IOP_{baseline}} \times 100\%,$$

$$SD_{IOPR\%} = \frac{SD_{IOPR}}{IOP_{baseline}} \times 100\%.$$

The second outcome measure was the proportion of eyes with complete success, which was defined as targeting end-point IOP (≤21mmHg or study-specific end-points) without anti-glaucoma medication. We assessed safety by considering the proportions of eyes with postoperative adverse events, including device malposition, device obstruction, nonobstructive peripheral anterior synechiae (PAS), hyphema, uveitis/iritis, macular



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edema, disc hemorrhage, best-corrected visual acuity (BCVA) loss > 2 lines, and elevated IOP ⩾ 10 mmHg over baseline.

**Risk of bias assessment**

The methodological quality of included studies was evaluated independently by two authors (RH and DG) based on the 6-item modified Jadad scale.[23-25] It evaluates the method of randomization, blinding, withdrawals and drop-outs, inclusion/exclusion criteria, adverse events, and the statistical analysis. According to these criteria, the scale scores range from 0 to 8 points. Studies with a score of 4 to 8 were deemed to be high quality, and studies with a score of 0 to 3 were deemed to be low quality. Any disagreement between raters was resolved through discussion with a third author (XW).

**Data extraction**

The studies' demographic details, participant characteristics, interventions, outcomes and limitations were independently extracted by two authors (RH, DG). If disagreements occurred, XW would check the data again.

**Statistical analyses**

Our analysis classified Hydrus implantation with phacoemulsification, 1-iStent implantation with phacoemulsification, 2-iStent implantation with phacoemulsification, and phacoemulsification alone as separate treatment nodes. The network meta-analysis was performed within a Bayesian framework using the Markov Chain Monte Carlo



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simulation implemented through the Aggregate Data Drug Information System software (ADDIS, V.1.16.6, Drugis, Groningen, NL).[26] We used a consistency model, which was based on 50,000 iterations for each 4 chains with a burn-in period of the first 20,000 iterations. Convergence was assessed using the Brooks-Gelman-Rubin method.[27] The measures of treatment effects were the weighted mean difference (WMD) for continuous outcomes and relative risk (RR) for dichotomous outcomes with 95% credible intervals (95% CI) based on the Bayesian statistical inference. Consistency analysis could be conducted in the presence of similarity and homogeneity, and on this basis, we could rank the effect of different treatment strategies. The higher ranking means the better the treatment is probably when considering treatment efficacy. It should be noted that even if the difference in the effect size between treatments is small, clinical decisions to choose treatment can still be guided by the probability of treatment rankings.[28,29] Inconsistency analysis was also conducted and similar results were obtained (supplementary tables 1-3).

Traditional pair-wise meta-analysis was performed using Stata 12.0 (Stata Corp., College Station, TX). We calculated and evaluated the odds ratio and its 95% CI of various study outcomes and merged data with inverse variance model. Forest plots were used to illustrate various study outcomes and their merged data (supplementary figures 1-3). A statistical evaluation of the heterogeneity of included RCTs was carried out using the $I^2$ parameter.

## RESULTS

### Literature search results



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The PRISMA flowchart of literature selection in this network meta-analysis is illustrated in figure 1. In total, 177 articles were retrieved through our initial search. After removal of duplicates and irrelevant studies (e.g., case reports, reviews or experimental studies), 95 articles remained for full-text review. Eighty-two articles were excluded on full-text review as they were not RCTs (43 studies), did not include interventions of interest (11 studies), or did not report outcomes of interest (28 studies). Full-text examinations excluded 7 additional articles as they described a RCT already included [subgroup analysis (3 studies), mid-term report (1 studies), or follow-up report after a trial ends (2 studies)], or did not report sufficient data (1 study). Finally, 6 articles (6 RCTs)[9,11-15] were included in this systematic review and network meta-analysis.

**Characteristics and outcomes of included studies**

Six studies comprising 1397 eyes with OAG and cataract were included in this meta-analysis. Figure 2 presents the network of eligible comparisons for the network meta-analysis. All included studies were prospective RCT investigations. The follow-up duration ranged from 12 months to 24 months and the patients' age distributions did not vary significantly among studies. The median sample size was 139 eyes (range 33 - 556). The main characteristics of the included studies are presented in table 1. The baseline characteristics of study populations of included RCTs are presented in table 2.

**Methodological quality of included studies**

Figure 3 presents the results of risk of bias assessment for each included study. The overall quality scores were good ranging from 5 to 8 (the modified-Jadad score >3). Of



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the 6 RCTs, appropriate randomization methods were described by 5 studies (83.3%); one study (16.7%) was described as double-blind, 3 studies (50%) were single-blind, and the remaining studies were not blinded (n = 2, 33.3%). Of the included studies, the majority provided an adequate description of withdrawals and dropouts (n = 5, 83.3%), inclusion/exclusion criteria (n = 6, 100%), the method used to assess adverse effects (n = 6, 100%), and the methods of statistical analysis (n = 6, 100%).

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**Table 1.** Study characteristics of included randomized controlled trials

| First author (year) | Location | Design | Glaucoma diagnosis | Interventions (No. of eyes) | Follow-up (month) |
|---|---|---|---|---|---|
| Fea et al. (2010)[9] | Italy | prospective double-masked | POAG | 1-iStent (n=12) Control (n=21) | 15 |
| Craven et al. (2012)[11] | United States | prospective open-label | mild or moderate OAG (POAG 91%) § | 1-iStent (n=98) Control (n=101) | 24 |
| Fernandez-Barrientos et al. (2010)[12] | Spain | prospective open-label | POAG or OHT | 2-iStent (n=17)† Control (n=16) | 12 |
| Samuelson et al. (2019)[13] | United States | prospective single-masked | mild to moderate POAG | 2-iStent (n=380)* Control (n=118) | 24 |
| Pfeiffer et al. (2015)[14] | multicenter | prospective single-masked | OAG (POAG 90%) OAG (POAG 82%) | Hydrus (n=44) Control (n=34) | 24 |
| Samuelson et al. (2019)[15] | multicenter | prospective single-masked | mild-to-moderate POAG | Hydrus (n=369) Control (n=187) | 24 |

1-iStent, 1-iStent implantation with phacoemulsification; 2-iStent, 2-iStent implantation with phacoemulsification; Hydrus, Hydrus implantation with phacoemulsification; Control, phacoemulsification alone; POAG, primary open-angle glaucoma; OAG, open-angle glaucoma; OHT, ocular hypertension. †2 first-generation stents; *second-generation iStent inject; § The data were presented based on the 1-year follow-up report of the same cohort.[10]



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**Table 2**. Baseline characteristics of study populations of included randomized controlled trials

| First author (year) | Interventions | Age | Female (%) | Ethnicity White (%) | MD (dB) | PSD (dB) | Medicated IOP | Washed-out IOP | No. of hypotensive medications |
|---|---|---|---|---|---|---|---|---|---|
| Fea et al. (2010)[9] | 1-iStent | 64.5±3.4 | 66.7 | - | - | - | 17.9±2.6 | - | 2.0±0.9 |
| | Control | 64.9±3.1 | - | - | - | - | 17.3±3.0 | - | 1.9±0.7 |
| Craven et al. (2012)[11] | 1-iStent | 74±8 | 59 § | 71 § | -3.8±3.0 | 2.9±1.8 | 18.6±3.4 | 25.4±3.6 | 1.6±0.8 |
| | Control | 73±9 | | | -3.9±3.6 | 2.8±1.9 | 17.9±3.0 | 25.2±3.6 | 1.5±0.6 |
| Fernandez-Barrientos et al. (2010)[12] | 2-iStent | 75.2±7.2 | 64.7 | - | - | - | - | 24.2±1.8 | 1.1±0.5 |
| | Control | 76.7±5.8 | 43.8 | - | - | - | - | 23.6±1.5 | 1.2±0.7 |
| Samuelson et al. (2019)[13] | 2-iStent | 69.0±8.2 | 58.1 | 72.9 | -3.4±3.3 | 3.5±2.5 | 17.5±3.0 | 24.8±3.3 | 1.6± 0.8 |
| | Control | 70.1±7.7 | 54.2 | 72.9 | -3.4±3.1 | 3.3±2.6 | 17.5±2.8 | 24.5±3.1 | 1.5± 0.7 |
| Pfeiffer et al. (2015)[14] | Hydrus | 72.8±6.6 | 60 | 96 | -5.6±5.4 | 5.1±4.6 | 18.9±3.3 | 26.3±4.4 | 2.0±1.0 |
| | Control | 71.5±6.9 | 42 | 98 | -8.4±7.8 | 5.2±4.3 | 18.6±3.8 | 26.6±4.2 | 2.0±1.1 |
| Samuelson et al. (2019)[15] | Hydrus | 71.1±7.9 | 55.8 | 78.9 | -3.6±2.5 | 3.2±2.2 | 17.9±3.1 | 25.5±3.0 | 1.7±0.9 |
| | Control | 71.2±7.6 | 56.1 | 81.8 | -3.6±2.6 | 3.1±1.9 | 18.1±3.1 | 25.4±2.9 | 1.7±0.9 |

MD, mean deviation; PSD, pattern standard deviation; 1-iStent, 1-iStent implantation with phacoemulsification; 2-iStent, 2-iStent implantation with phacoemulsification; Hydrus, Hydrus implantation with phacoemulsification; Control, phacoemulsification alone; IOP, intraocular pressure. Results are mean ± standard deviation unless noted. § The data were presented based on the 1-year follow-up report of the same cohort.[10]



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**Efficacy**

Table 3 and 4 present the WMD and 95% CI of IOPR and IOPR%, respectively for all possible comparisons by the network meta-analysis using the consistency model. In terms of the IOPR, the 2-iStent group (1.88, 95% CI 0.41 to 3.62 mmHg) and the Hydrus group (2.21, 95% CI 0.52 to 3.71 mmHg) showed greater IOP reduction than the control group at follow-up end. In terms of the IOPR%, the 1-iStent group (7.65%, 95% CI 0.66 to 15.45%), the 2-iStent group (7.31%, 95% CI 1.14 to 14.60%), and the Hydrus group (8.66%, 95% CI 2.02 to 14.81%) all showed greater IOP reduction than the control group. There was no significant difference in IOPR and IOPR% among the 1-iStent group, the 2-iStent group, and the Hydrus group. Figure 4 presents the rank probabilities of these groups for the treatment efficacy in terms of the IOPR and the IOPR%. Overall, the Hydrus group showed a higher probability to be the best treatment in lowering IOP than the 1-iStent group and the 2-iStent group (0.60 vs. 0.11, 0.60 vs. 0.29 in IOPR; 0.46 vs. 0.31, 0.46 vs. 0.23 in IOPR%, respectively).

**Table 3**. Network meta-analysis results in IOP reduction at follow-up end

| Weighted mean difference (95% CI), mmHg | | | |
|---|---|---|---|
| 1-iStent | | | |
| -0.47 (-2.86, 1.65) | 2-iStent | | |
| -0.81 (-3.02, 1.64) | -0.35 (-2.34, 2.25) | Hydrus | |
| 1.40 (-0.19, 3.06) | 1.88 (0.41, 3.62)* | 2.21 (0.52, 3.71)* | Control |

*Statistical significance. IOP, intraocular pressure; CI, credible interval; 1-iStent, 1-iStent implantation with phacoemulsification; 2-iStent, 2-iStent implantation with phacoemulsification; Hydrus, Hydrus implantation with phacoemulsification; Control, phacoemulsification alone.



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**Table 4**. Network meta-analysis results in the percentage of IOP reduction at follow-up end

| Weighted mean difference (95% CI), % | | | |
|---|---|---|---|
| 1-iStent | | | |
| 0.32 (-10.02, 10.27) | 2-iStent | | |
| -1.08 (-10.31, 9.24) | -1.33 (-9.79, 8.75) | Hydrus | |
| 7.65 (0.66, 15.45)* | 7.31 (1.14, 14.60)* | 8.66 (2.02, 14.81)* | Control |

*Statistical significance. IOP, intraocular pressure; CI, credible interval; 1-iStent, 1-iStent implantation with phacoemulsification; 2-iStent, 2-iStent implantation with phacoemulsification; Hydrus, Hydrus implantation with phacoemulsification; Control, phacoemulsification alone.

Table 5 presents the RR and 95% CI of the proportions of eyes with complete success for all possible comparisons by the network meta-analysis using the consistency model. The 1-iStent group, the 2-iStent group, and the Hydrus group showed a likely positive effect in the proportions of complete success versus the control group, however, the differences were not statistically significant. There was no significant difference in the proportion of complete success among the 1-iStent group, the 2-iStent group, and the Hydrus group. Figure 4 presents the rank probability of these groups for treatment effect in terms of the proportion of complete success. The 2-iStent group and the Hydrus group showed the higher probability to achieve the complete success than the 1-iStent group and the control group. The 2-iStent group and the Hydrus group showed the close probability to be the best treatment (0.44 and 0.40, respectively).



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| **Table 5**. Network meta-analysis results in medication free at follow-up end | | | |
|---|---|---|---|
| Relative risk (95% CI) in proportions of complete success | | | |
| 1-iStent | | | |
| 0.60 (0.01, 10.07) | 2-iStent | | |
| 0.63 (0.04, 16.37) | 1.05 (0.08, 68.13) | Hydrus | |
| 2.65 (0.37, 29.48) | 4.33 (0.79, 145.22) | 4.06 (0.51, 38.62) | Control |

Complete success, targeting end-point intraocular pressure (⩽21mmHg or study-specific end-points) without anti-glaucoma medication. CI, credible interval; 1-iStent, 1-iStent implantation with phacoemulsification; 2-iStent, 2-iStent implantation with phacoemulsification; Hydrus, Hydrus implantation with phacoemulsification; Control, phacoemulsification alone.

**Adverse events**

Main adverse events of included studies are present in table 6. In summary, device malposition occurred in 3.9% of 1-iStent eyes, 1.4% of 2-iStent eyes, and was not reported in Hydrus eyes. Device obstruction occurred in 3.9% of 1-iStent eyes, 6.0% of 2-iStent eyes, and 3.3% of Hydrus eyes. Focal PAS, which was considered as nonobstruction, occurred in 1.7% of 2-iStent eyes, 15.3% of Hydrus eyes, and was not reported in 1-iStent eyes.

There were few sight-threatening complications in included studies. At follow-up end, 2.5% of 2-iStent eyes and 1.2% of Hydrus eyes had lost >2 lines compared with preoperative BCVA, however, the proportions were numerically less in treatment eyes than control eyes (3.7% and 1.7%, respectively in 2-iStent and Hydrus studies). Macular edema occurred in 0.8% of 1-iStent eyes and 2.4% of Hydrus eyes, which were also less in treatment eyes than control eyes (1.4% and 2.5%, respectively in 1-iStent and Hydrus studies). Hyphema, disc hemorrhage, and IOP elevation ⩾10 mmHg over baseline occurred at a rate of 2% or less in all treatment eyes of included studies. A higher



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proportion of postoperative uveitis/iritis was observed in 2-iStent eyes and Hydrus eyes versus control eyes (5.5% vs. 3.7% in 2-iStent studies; 5.0% vs. 2.9% in Hydrus studies), while a lower proportion of uveitis/iritis was reported in 1-iStent eyes versus control eyes (0.8% vs. 4.3%).

**Table 6**. Main adverse events after Hydrus and iStent implantation combined with phacoemulsification

| Adverse events | 1-iStent | 1-iStent control | 2-iStent | 2-iStent control | Hydrus | Hydrus control |
|---|---|---|---|---|---|---|
| Number of eyes | 128 | 138 | 403 | 135 | 417 | 236 |
| Device malposition | 3.9% | NA | 1.4% | NA | NR | NA |
| Device obstruction | 3.9% | NA | 6.0% | NA | 3.3% | NA |
| Focal PAS, nonobstructive | NR | NR | 1.7% | NR | 15.3% | 2.1% |
| Hyphema | NR | NR | 2.0% | NR | 0.4% | 0.4% |
| Uveitis/iritis | 0.8% | 4.3% | 5.5% | 3.7% | 5.0% | 2.9% |
| Macular edema | 0.8% | 1.4% | NR | NR | 2.4% | 2.5% |
| Disc hemorrhage | 0.8% | 2.2% | NR | NR | 0.2% | NR |
| BCVA loss > 2 lines | NR | NR | 2.5% | 3.7% | 1.2% | 1.7% |
| Elevated IOP ≥ 10 mmHg over baseline | NR | NR | 2.0% | 0.7% | 0.9% | 3.0% |

NA, not applicable; NR, not reported; PAS, peripheral anterior synechiae; BCVA, best-corrected visual acuity; IOP, intraocular pressure; 1-iStent, 1-iStent implantation with phacoemulsification; 2-iStent, 2-iStent implantation with phacoemulsification; Hydrus, Hydrus implantation with phacoemulsification; 1-iStent control, 2-iStent control, and Hydrus control, phacoemulsification alone as the control for each treatment.



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## DISCUSSION

To our knowledge, this is the first study to indirectly compare the efficacy and safety of two commercially available Schlemm's canal-based MIGS devices for glaucoma management with concurrent cataract surgery using published RCT data. Based on the network meta-analysis, the Hydrus and 2-iStent implantation combined with phacoemulsification achieved statistically greater reductions in IOP than phacoemulsification alone, in terms of either the absolute IOP reduction values or the percentage. Although no significant difference was found among the three groups, the Hydrus device may provide better IOP reduction effect than the 1- and 2-iStent device based on the rank probability of the existing data. In terms of the proportion of complete success, no statistically significant difference was found among the 1-iStent group, the 2-iStent group, the Hydrus group, and the phacoemulsification alone group, while the rank probability analysis shows the Hydrus and the 2-iStent implantation may achieve the complete success at a higher probability versus the 1-iStent implantation and phacoemulsification alone.

Previous laboratory studies compared the Hydrus and the iStent devices using human cadaveric anterior segment perfusion models.[30,31] Hays et al showed the Hydrus increased the aqueous outflow facility by a greater percentage (73% vs. 34%) and at a wider range of perfusion pressures (20 to 50 mm Hg vs. 40 mm Hg) than the 2 first-generation stents.[30] Similar results were shown in a recent randomized paired comparison study by Toris et al, in which Hydrus eyes had higher increase in the outflow facility than double iStent inject eyes (79% vs. 11%).[31] These in-vitro laboratory findings support our meta-



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analysis results that the Hydrus device may lower the IOP better than the 2-iStent device. Furthermore, the negative correlation between the Schlemm's canal area and the IOP has been observed in vivo in OAG eyes, and normal eyes have larger Schlemm's canal area than OAG eyes.[32] Trabecular meshwork stretch is also one of mechanisms of the pilocarpine treatment effect.[33] It is therefore likely that the trabecular meshwork stretch and Schlemm's canal dilation by the 8-mm scaffold segment of the Hydrus help to provide additional benefits to lower the outflow resistance to collector channels, and accordingly achieve greater reduction in IOP versus the iStent.[30,31]

Our results are also in line with the conclusion of the COMPARE study, which was a prospective, multicenter RCT to compare the Hydrus and the 2-iStent for standalone treatment.[34] One hundred fifty-two eyes from 152 OAG patients were randomized to either Hydrus or 2 first-generation stents implantation and followed for 12 months. Both groups had significantly reduced medication use at 12 months compared with the baseline, and the reduction in the Hydrus group was significantly greater by 0.6 medication than the 2-iStent group. More Hydrus patients were medication free at the follow-up end (46.6% vs. 24.0%). While it is difficult to directly compare the present meta-analysis results of combined surgery with the standalone treatment results, both studies suggest the Hydrus might be more effective to treat OAG versus the 2-iStent.

Two recent retrospective studies compared the real-world outcomes of the Hydrus and the iStent combined with phacoemulsification. For the iStent group, Lee et al[35] included patients implanted with one first-generation iStent, and Holmes et al[36] included patients



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implanted with second-generation iStent inject. Both studies showed sustained IOP reduction with a good safety profile for the MIGS groups during the 2-year follow-up and no significant difference was found between the Hydrus and the iStent. For medication use, Lee et al[35] reported a 0.5-medication reduction advantage with the Hydrus group compared to the iStent group, while Holmes et al[36] reported an additional reduction with the iStent inject group by 0.5 medication on average compared to the Hydrus group. Of note, in the latter study, the Hydrus group had significantly higher IOP, more glaucoma medications, and greater visual field damage at baseline than the iStent inject group, which may potentially bias the results. The baseline visual field mean deviation of Hydrus group in Holmes et al's study was also seemingly greater than that of our included RCTs (-8.8 dB vs. -5.6 dB[14] / -3.6 dB[15] on average), and their findings are somewhat different to our study. It is unclear whether these differences in efficacy comparison are because of the difference in glaucoma severity of study populations.

The theoretical basis for incremental efficacy with implantation of multiple trabecular stents was established previously based on in vitro perfusion models.[37-39] Several clinical studies have shown that use of additional first-generation iStent may result in greater IOP reduction with reduced medication use versus a single implant.[40-42] One study by Hooshmand et al[18] did not demonstrate a statistically significant difference between a single iStent and double iStent inject during one-year follow-up, and earlier recommencement of medications for optimal IOP control was reported with the iStent inject. In the present study, the majority of study participants of the 2-iStent group were implanted with the iStent inject. Though there was no significant difference between the



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1-iStent group and the 2-iStent group across our outcome measures, the 2-iStent group showed statistically significant reduction in absolute values of IOPR versus phacoemulsification alone, while the 1-iStent group not. Furthermore, the rank probability analysis shows the 2-iStent implantation may achieve complete success at a higher probability than the 1-iStent implantation (0.44 vs. 0.16, respectively). Larger population size and longer term follow-up might be needed to obtain more definitive results for the efficacy comparison between a single first-generation iStent and second-generation iStent inject.

The overall safety profiles of three treatment groups were favorable and few serious ocular adverse events were reported in included RCTs. Device obstruction, which may be caused by pigment, iris tissue, or hyphema, occurred in either device (3.9% of 1-iStent eyes, 6.0% of 2-iStent eyes, and 3.3% of Hydrus eyes). Device malposition was observed in 3.9% of 1-iStent eyes and 1.4% of 2-iStent eyes, while not reported in the Hydrus eyes. A likely explanation is the Hydrus was designed with a longer segment residing within the lumen of Schlemm's canal versus the iStent, which allows a better positioning stability. PAS was more frequently observed in the Hydrus eyes versus the 1-iStent and 2-iStent eyes (15.3%, 0%, and 1.7%, respectively). Our result is consistent with the COMPARE study, which indicates that the PAS formation was more common in the Hydrus group.[34] It is noteworthy that subgroup analyses of included RCTs did not show the significant difference in the IOP reduction between eyes with PAS and without PAS.[14,15] Therefore, most of the PAS might not be obstructive in eyes treated with



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Hydrus implantation. Nevertheless, caution should be taken for the PAS as it may eventually result in the device obstruction in the long term.

The baseline characteristics of study populations were overall similar in included RCTs. A common preoperative diagnosis was OAG, including 3 studies only with primary open-angle glaucoma,[9,13,15] one study with ocular hypertension (11.8%),[12] one study with pseudoexfoliative glaucoma (6%) and pigmentary glaucoma (3%),[11] and one study with pseudoexfoliative glaucoma (10%).[14] Angle-closure, traumatic, uveitic, or neovascular glaucoma, or history of incisional glaucoma surgery were all excluded. Baseline visual field damage was comparable in included RCTs, except one Hydrus study with greater mean deviation and pattern standard deviation.[14] Real-world data show that the Hydrus have been used in patients with more advanced disease versus the iStent inject, which are in line with the present included RCT.[36] As shown in table 2, the majority of study populations were Caucasian. There was no significant difference in age and sex among groups and these demographics characteristics were well matched for the treatment and control groups in each study.

The surgical techniques of the Hydrus and the iStent are similar which both require a clear view of the angle structure, and the devices are introduced into the anterior chamber through a clear corneal incision and implanted through the trabecular meshwork.[34] Despite the increasing use of MIGS in clinical practice, data to compare the cost-effectiveness of these devices remain limited.[43] Separate studies have shown either the Hydrus[44] or the iStent inject[45,46] combined with phacoemulsification is cost-effective for



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patients with mild to moderate OAG versus phacoemulsification alone. A recent study by Sood et al[47] compared the cost-effectiveness of the Hydrus and the iStent inject in combination with phacoemulsification for patients with mild to moderate primary open-angle glaucoma. Both devices were not cost-effective versus phacoemulsification alone over the first 2 years due to the initial costs of the surgeries, however, these devices were demonstrated to be cost-effective treatment options over a lifetime horizon based on the simulation analysis. In comparison, the Hydrus was slightly more cost-effective than the iStent inject, and the authors concluded it was likely due to the slightly better treatment efficacy associated with the Hydrus versus the iStent inject at the identical costs.

Although the results of this meta-analysis may be important for treatment considerations, there are several imitations that should be mentioned. First, we included prospective high-quality RCTs, however, the number of studies is relatively small and the sample size of included studies varied ranging from 33 to 556 eyes, which may potentially bias the results. Further research is necessary to replenish this meta-analysis to offer more convincing conclusion. Second, the details of adverse events were not always reported in each study, and due to the limited data, we did not perform statistical analysis to compare the difference in treatment safety among groups. Third, for the 2-iStent group, we merged the data analyzing 2 first-generation stents[12] and second-generation iStent inject,[13] which is preloaded with 2 stents. It is unclear whether there is difference in efficacy between 2 first-generation stents and double iStent inject, as few studies have been conducted to compare these 2 stents. In vitro perfusion model study did not demonstrate significant difference in outflow facility increase between the 2 stents.[31] A subgroup analysis by



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Salimi et al[48] shows no significant difference was found between the 2 generations in terms of IOP reduction and medication use in eyes with primary angle closure glaucoma.

## CONCLUSION

Based on the existing data at present, our network meta-analysis results indicate that the Hydrus device might be slightly more effective in combination with phacoemulsification to treat OAG versus the 1- or 2-iStent device, and these devices have overall good safety profiles. Our findings may aid clinicians in selecting appropriate treatments in glaucoma management, particularly for patients with mild-to-moderate glaucoma and cataract, however, there might be of some uncertainty due to the limited included data. Further prospective, large-scale RCTs that directly compare these MIGS devices combined with cataract surgery are needed to investigate whether our findings are robust.

**Acknowledgements** The authors thank the researchers whose studies were included in this network meta-analysis and provided useful data to us.

**Contributors** All authors made a substantial contribution to this work. RH: study conception, study design, data collection, data analysis, manuscript writing and final manuscript approval; DG: study design, data collection and critical manuscript revision; NH: statistical analysis; XX: statistical analysis; XW: study conception, study design, critical manuscript revision and final manuscript approval.



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**Funding** This study was supported by the National Natural Science Foundation of China Grant 81900854, the Zhejiang Provincial Natural Science Foundation of China Grant LQ17H120003, and the Health and Family Planning Commission of Zhejiang Province Grant 2016KYA089.

**Competing interests** None declared.

**Patient consent** Not required.

**Ethics approval** This study was approved by the ethics committee of The First Affiliated Hospital, Zhejiang University School of Medicine and followed the tenets of the Declaration of Helsinki (Approval No. 20191003).

**Provenance and peer review** Not commissioned; externally peer reviewed.

**Data sharing statement** Extra data are available from the first author on request.

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## Figure legends

**Figure. 1.** Flow chart of study selection process. RCT, randomized controlled trial.

**Figure. 2.** Network graph of all treatment comparisons of included studies. Each node represents a treatment strategy. Lines represent direct comparisons within the randomized controlled trials.

**Figure. 3.** Modified-Jadad scores for risk of bias assessment. NR, not reported; NA, not applicable.

**Figure. 4.** Rank probability of treatments based on the absolute value of intraocular pressure (IOP) reduction, the percentage of IOP reduction, and the proportion of complete success. 1-iStent, 1-iStent implantation with phacoemulsification; 2-iStent, 2-iStent implantation with phacoemulsification; Hydrus, Hydrus implantation with phacoemulsification; Control, phacoemulsification alone. Rank 1 is best and Rank 4 is worst.

**Supplementary Figure. 1.** Forest plots for results from traditional pairwise meta-analysis for the absolute value of intraocular pressure reduction: (A) 1-iStent group; (B) 2- iStent group; (C) Hydrus group. SMD, standardized mean difference; CI, credible interval.

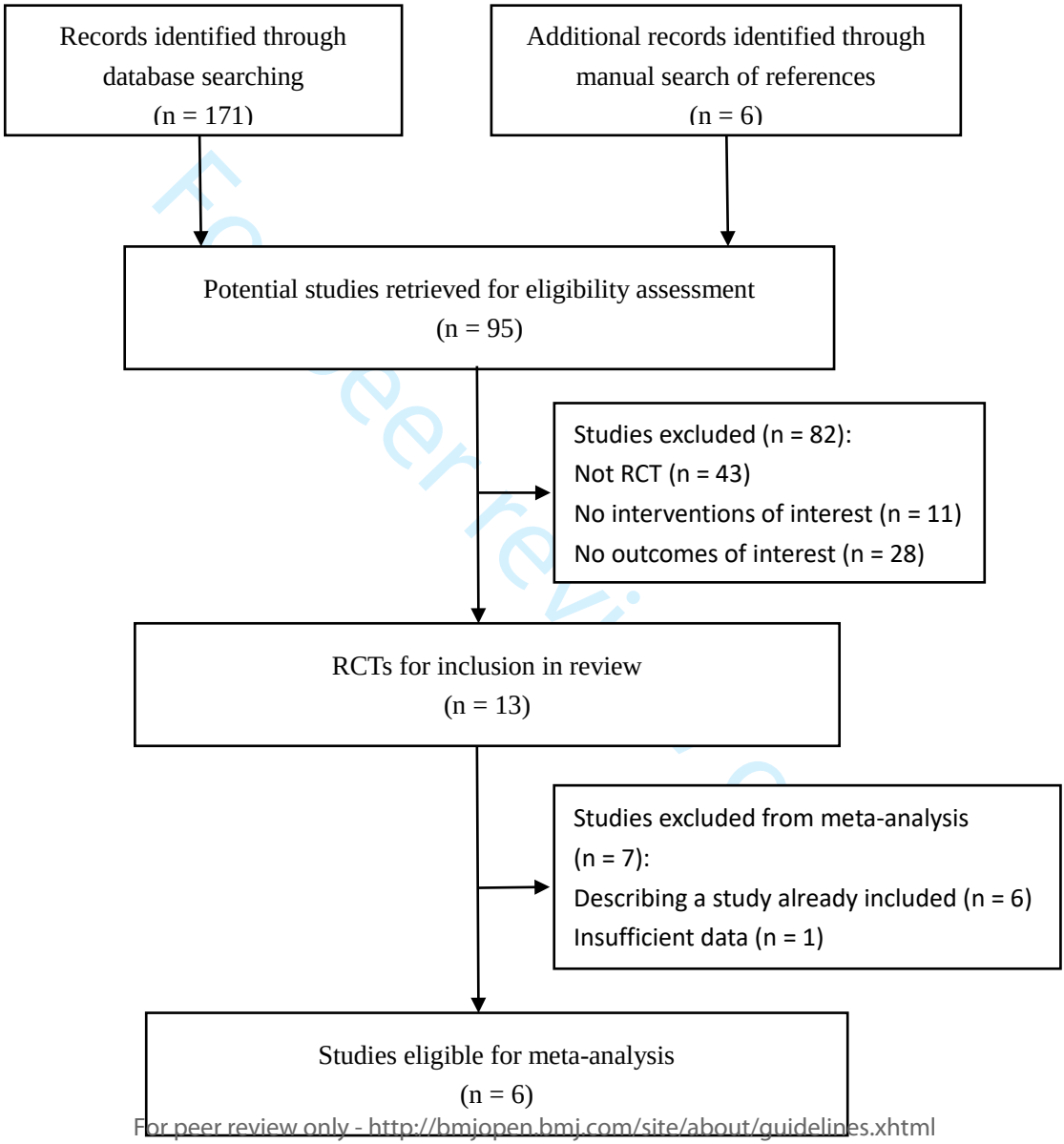
**Supplementary Figure. 2.** Forest plots for results from traditional pairwise meta-analysis for the percentage of intraocular pressure reduction: (A) 1-iStent group; (B) 2-iStent group; (C) Hydrus group. SMD, standardized mean difference; CI, credible interval.

**Supplementary Figure. 3.** Forest plots for results from traditional pairwise meta-analysis for the proportion of complete success: (A) 1-iStent group; (B) 2- iStent group; (C) Hydrus group. RR, relative risk; CI, credible interval.

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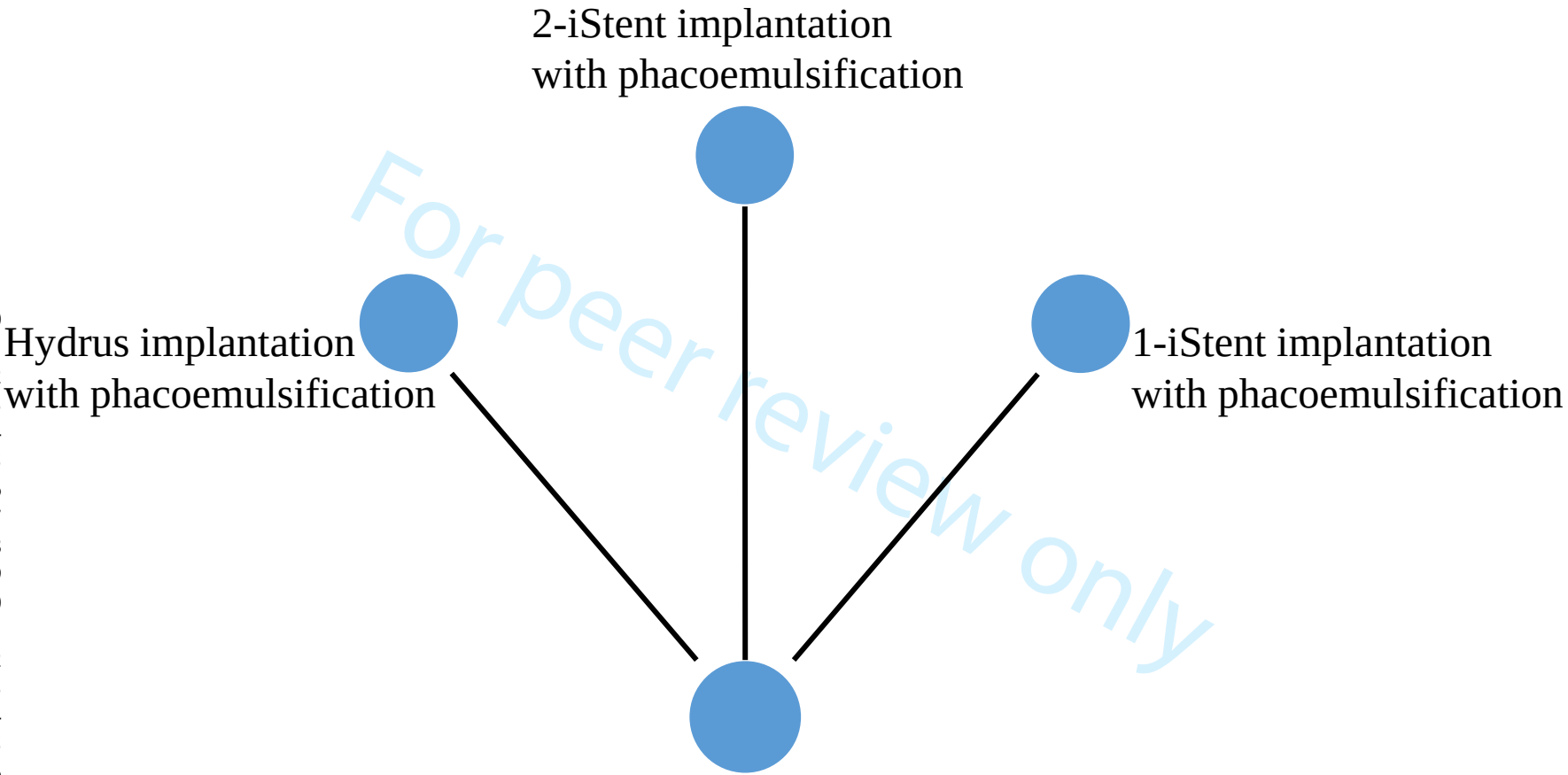


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Fig. 1




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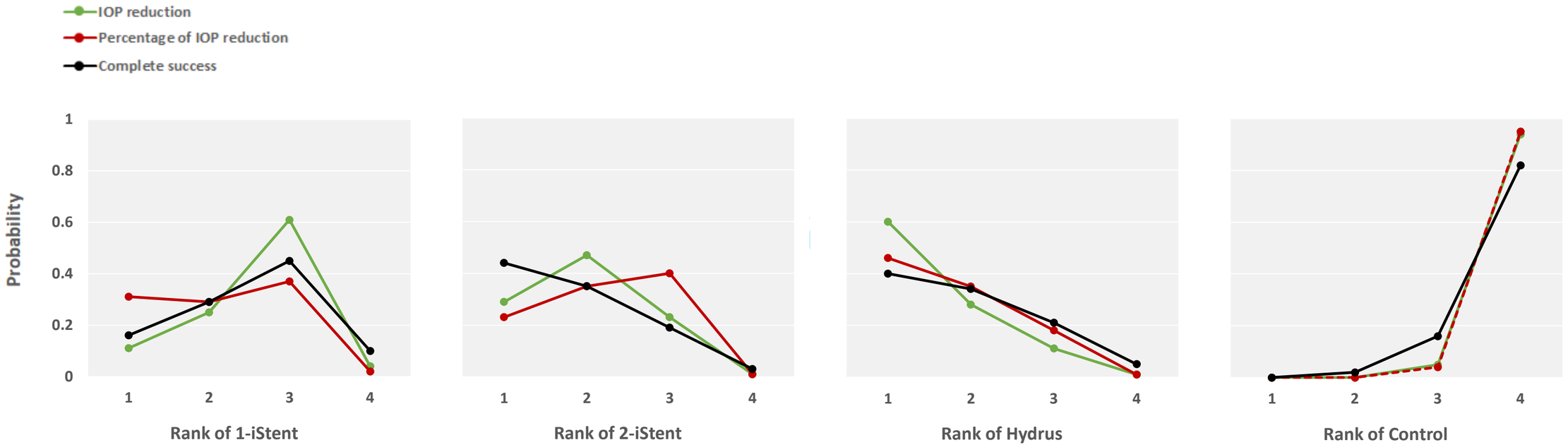
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Fig. 2

| | Randomized study | Randomization method appropriate | Double blind study | Blinding method appropriate | Description of withdrawals and dropouts | Description of inclusion/exclusion criteria | Method to assess adverse effects described | Methods of statistical analysis described | TOTAL |
|---|---|---|---|---|---|---|---|---|---|
| Fea et al. (2010)[9] | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 8 |
| Craven et al. (2012)[11] | 1 | NR | 0 | NA | 1 | 1 | 1 | 1 | 5 |
| Fernandez-Barrientos et al. (2010) [12] | 1 | 1 | 0 | NA | NR | 1 | 1 | 1 | 5 |
| Samuelson et al. (2019)[13] | 1 | 1 | 0 | NA | 1 | 1 | 1 | 1 | 6 |
| Pfeiffer et al. (2015)[14] | 1 | 1 | 0 | NA | 1 | 1 | 1 | 1 | 6 |
| Samuelson et al. (2019)[15] | 1 | 1 | 0 | NA | 1 | 1 | 1 | 1 | 6 |

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Fig. 3


IOP reduction
Percentage of IOP reduction
Complete success
Probability
1
0.8
0.6
0.4
0.2
0
1
2
3
4
Rank of 1-iStent
Rank of 2-iStent
Rank of Hydrus
Rank of Control

**Supplementary Table 1**. Network meta-analysis results in IOP reduction at follow-up end (Inconsistency Model).

| Weighted mean difference (95% CI), mmHg | | | |
|---|---|---|---|
| 1-iStent | | | |
| -0.46 (-2.81, 1.75) | 2-iStent | | |
| -0.81 (-2.98, 1.61) | -0.33 (-2.47, 2.13) | Hydrus | |
| 1.44 (-0.13, 3.19) | 1.90 (0.43, 3.67)* | 2.23 (0.62, 3.77)* | Control |

*Statistical significance. IOP, intraocular pressure; CI, credible interval; 1-iStent, 1-iStent implantation with phacoemulsification; 2-iStent, 2-iStent implantation with phacoemulsification; Hydrus, Hydrus implantation with phacoemulsification; Control, phacoemulsification alone.

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**Supplementary Table 2**. Network meta-analysis results in the percentage of IOP reduction at follow-up end (Inconsistency Model).

| Weighted mean difference (95% CI), % | | | |
|---|---|---|---|
| 1-iStent | | | |
| 0.62 (-9.89, 9.96) | 2-iStent | | |
| -1.03 (-10.66, 9.24) | -1.62 (-9.84, 8.69) | Hydrus | |
| 7.82 (0.21, 15.43)* | 7.19 (1.27, 14.75)* | 8.80 (2.10, 14.92)* | Control |

*Statistical significance. IOP, intraocular pressure; CI, credible interval; 1-iStent, 1-iStent implantation with phacoemulsification; 2-iStent, 2-iStent implantation with phacoemulsification; Hydrus, Hydrus implantation with phacoemulsification; Control, phacoemulsification alone.

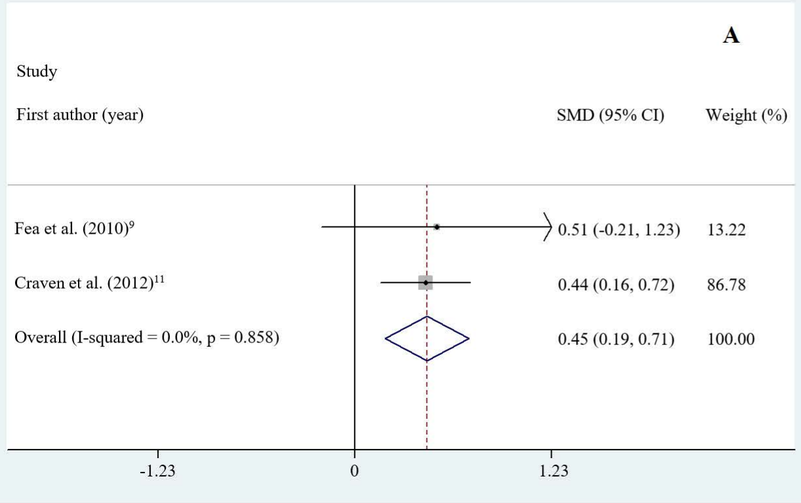
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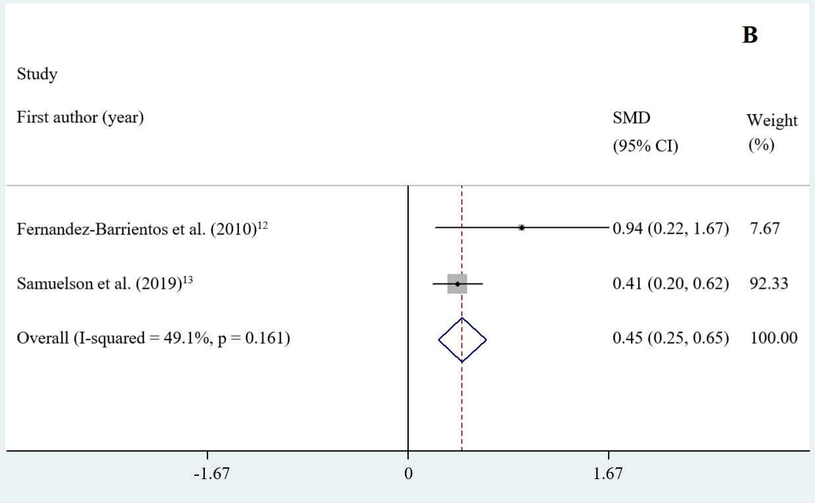
**Supplementary Table 3**. Network meta-analysis results in medication free at follow-up end (Inconsistency Model).

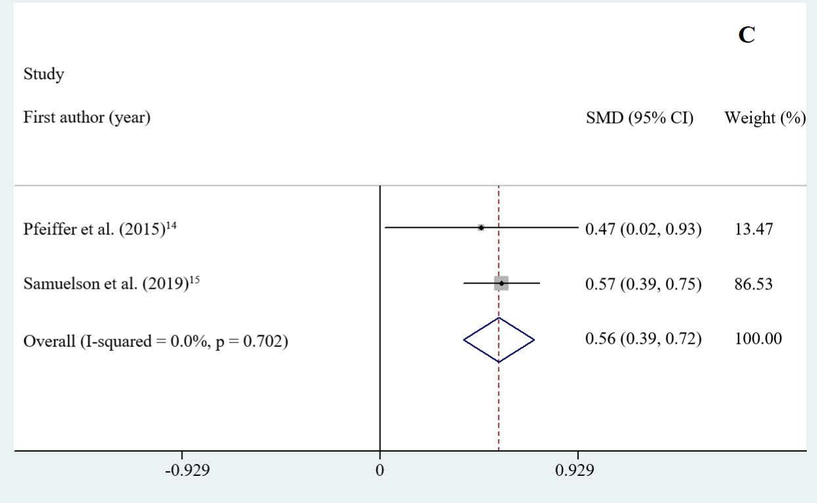
| Relative risk (95% CI) in proportions of complete success | | | |
|---|---|---|---|
| 1-iStent | | | |
| 0.59 (0.01, 10.61) | 2-iStent | | |
| 0.66 (0.04, 18.08) | 1.15 (0.08, 84.20) | Hydrus | |
| 2.63 (0.36, 31.82) | 4.44 (0.76, 139.89) | 3.98 (0.45, 36.31) | Control |

CI, credible interval; complete success, targeting end-point intraocular pressure (⩽21mmHg or study-specific end-points) without anti-glaucoma medication; 1-iStent, 1-iStent implantation with phacoemulsification; 2-iStent, 2-iStent implantation with phacoemulsification; Hydrus, Hydrus implantation with phacoemulsification; Control, phacoemulsification alone.

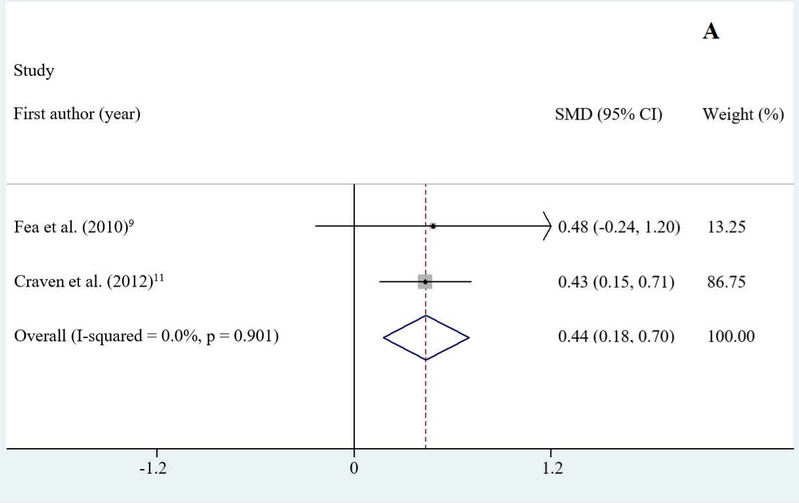
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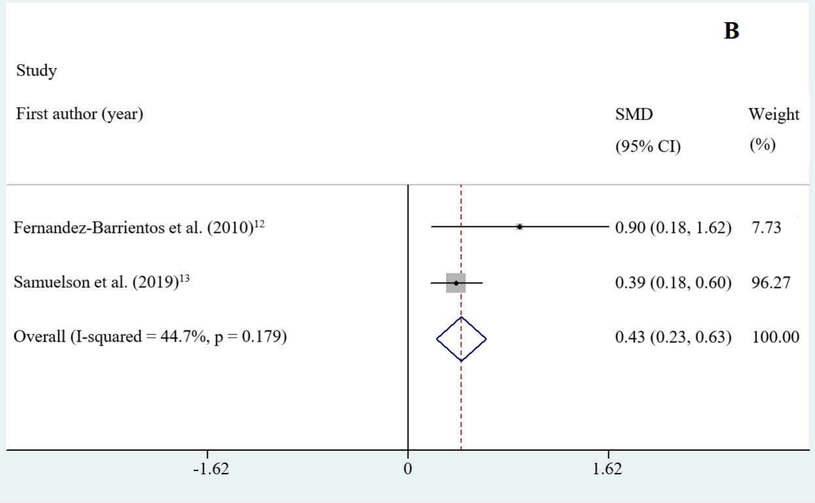


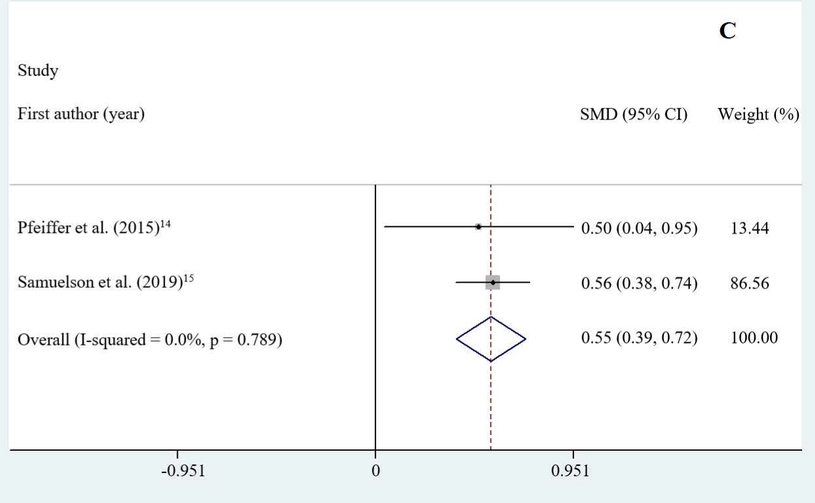




Supplementary figure 1

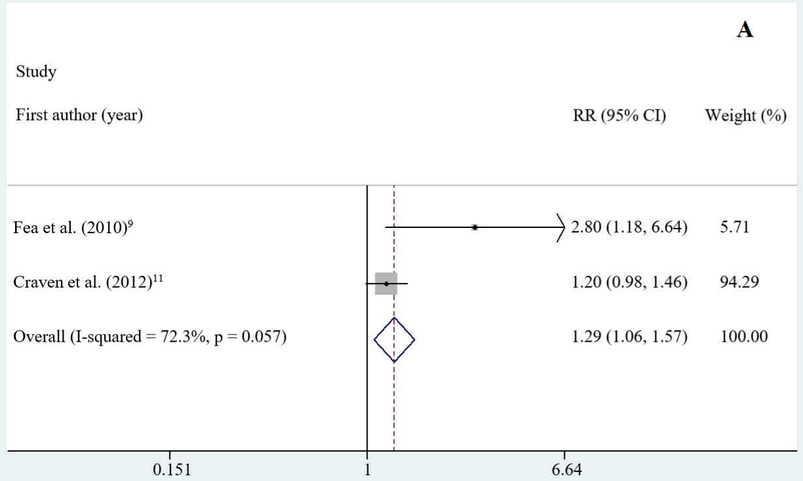


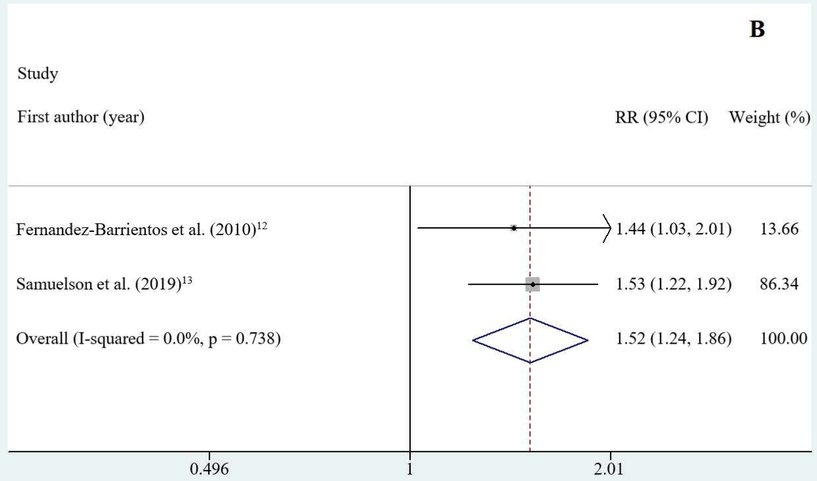



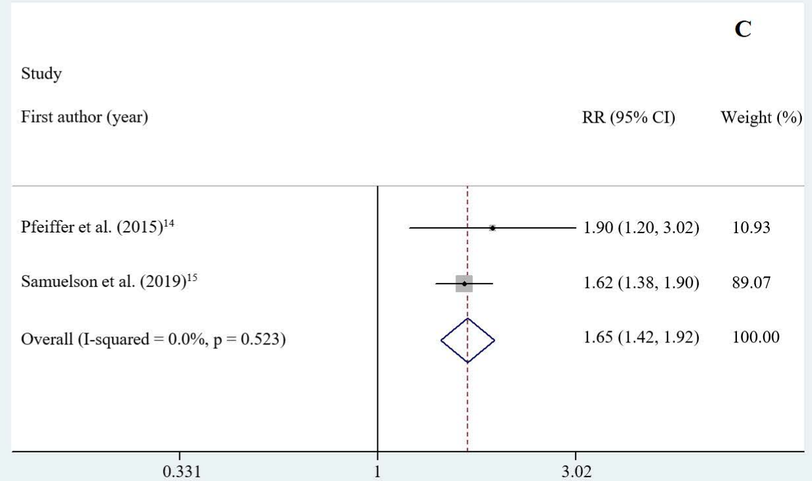




Supplementary figure 2

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Supplementary figure 3

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## Supplementary Appendix 1. Search strategies

We searched PubMed, Web of Science, Cochrane Library and ClinicalTrials.gov to identify relevant studies by the end of Sep 2020. We provided below the search strategy of PubMed database as an example.

#1 iStent

#2 trabecular micro-bypass stent

#3 trabecular bypass

#4 Hydrus

#5 Hydrus microstent

#6 schlemm canal microstent

#7 #1 OR #2 OR #3 OR #4 OR #5 OR #6

#8 cataract

#9 cataract surgery

#10 phacoemulsification

#11 #8 OR #9 OR #10

#12 #7 AND #11

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# PRISMA 2020 Checklist

| Section and Topic | Item # | Checklist item | Location where item is reported |
|---|---|---|---|
| **TITLE** | | | |
| Title | 1 | Identify the report as a systematic review. | P1 |
| **ABSTRACT** | | | |
| Abstract | 2 | See the PRISMA 2020 for Abstracts checklist. | P2-3 |
| **INTRODUCTION** | | | |
| Rationale | 3 | Describe the rationale for the review in the context of existing knowledge. | P4 |
| Objectives | 4 | Provide an explicit statement of the objective(s) or question(s) the review addresses. | P5 |
| **METHODS** | | | |
| Eligibility criteria | 5 | Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses. | P6-7, P8 |
| Information sources | 6 | Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted. | P6 |
| Search strategy | 7 | Present the full search strategies for all databases, registers and websites, including any filters and limits used. | P6, Supplementary Appendix 1 |
| Selection process | 8 | Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process. | P6 |
| Data collection process | 9 | Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process. | P8 |
| Data items | 10a | List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect. | P7-8 |
| | 10b | List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information. | P7-8 |
| Study risk of bias assessment | 11 | Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process. | P8 |
| Effect measures | 12 | Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results. | P8-9 |
| Synthesis methods | 13a | Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)). | P8 Table 1 |
| | 13b | Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions. | P8-9 |
| | 13c | Describe any methods used to tabulate or visually display results of individual studies and syntheses. | P9 |
| | 13d | Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used. | P8-9 |

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| | | | |
|---|---|---|---|
| | 13e | Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression). | P9 |
| | 13f | Describe any sensitivity analyses conducted to assess robustness of the synthesized results. | P9 |
| Reporting bias assessment | 14 | Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases). | |
| Certainty assessment | 15 | Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome. | P9 |



## PRISMA 2020 Checklist

| Section and Topic | Item # | Checklist item | Location where item is reported |
|---|---|---|---|
| **RESULTS** | | | |
| Study selection | 16a | Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram. | P10 Figure 1 |
| | 16b | Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded. | P10 |
| Study characteristics | 17 | Cite each included study and present its characteristics. | P10 Table 1, 2 |
| Risk of bias in studies | 18 | Present assessments of risk of bias for each included study. | P10-11 Figure 3 |
| Results of individual studies | 19 | For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots. | P14-17 |
| Results of syntheses | 20a | For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies. | P14-16 |
| | 20b | Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect. | P14-17 |
| | 20c | Present results of all investigations of possible causes of heterogeneity among study results. | Tables 1,2 |
| | 20d | Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results. | P9 Supplementary tables 1-3 |
| Reporting biases | 21 | Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed. | |
| Certainty of evidence | 22 | Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed. | P14-16 |
| **DISCUSSION** | | | |
| Discussion | 23a | Provide a general interpretation of the results in the context of other evidence. | P18-21 |
| | 23b | Discuss any limitations of the evidence included in the review. | P23 |
| | 23c | Discuss any limitations of the review processes used. | P23-24 |
| | 23d | Discuss implications of the results for practice, policy, and future research. | P24 |

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| | | | |
|---|---|---|---|
| **OTHER INFORMATION** | | | |
| Registration and protocol | 24a | Provide registration information for the review, including register name and registration number, or state that the review was not registered. | |
| | 24b | Indicate where the review protocol can be accessed, or state that a protocol was not prepared. | P5 |
| | 24c | Describe and explain any amendments to information provided at registration or in the protocol. | |
| Support | 25 | Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review. | P25 |
| Competing interests | 26 | Declare any competing interests of review authors. | P25 |
| Availability of data, code and other materials | 27 | Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review. | P25 |

*From:* Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

For more information, visit: http://www.prisma-statement.org/

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